

Design, Modelling and Performance Evaluation of Solid-State Transformer (SST)

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in partial fulfilment of the requirements for the award of the degree of

Bachelor of Technology

in

Electrical Engineering



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Declaration

We declare that this Report represents our work in our own words. Wherever the ideas or words of others have been used, the corresponding sources have been properly acknowledged and referenced. We further declare that the work entitled “**Design, Modelling and Performance Evaluation of Solid-State Transformer (SST)**” was carried out under the supervision of **Dr. Ward Ul Hijaz Paul** and has not been submitted, in full or in part, to any other university or institution for the award of a degree or diploma.

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Certificate of Approval

The Report entitled “**Design, Modelling and Performance Evaluation of Solid-State Transformer (SST)**”, submitted by Afas Yousuf, Malik Inam-ul-Haq, Najmu Saqib, and Saif Ul Fayaz under the guidance of Dr. Ward Ul Hijaz Paul, is hereby approved as a satisfactory study in Electrical Engineering and is accepted in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology from the Institute of Technology, University of Kashmir, Zakura Campus, Srinagar, India.

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Abstract

The Solid-State Transformer (SST) has emerged as a promising alternative to conventional power transformers due to its compact size, high efficiency, electrical isolation, and improved power quality. This Report presents the design, modelling, simulation, hardware implementation, and performance evaluation of a 50 W single-phase Solid-State Transformer employing a three-stage power conversion architecture. The proposed system consists of a single-phase AC–DC full-bridge rectifier, an isolated DC–DC converter incorporating a 1:1 high-frequency ferrite-core transformer, and a DC–AC inverter controlled using Sinusoidal Pulse Width Modulation (SPWM).

A detailed MATLAB/Simulink model was developed to analyse the steady-state performance of the proposed SST. A preliminary laboratory-scale hardware prototype operating from a 15.5 V AC input supply was also developed using IRF3205 MOSFETs, IR2110 gate driver ICs, an Arduino Mega 2560 microcontroller, and an LC output filter to support future instrumented validation.

The proposed SST achieved stage-wise efficiencies of 95.01%, 98.39%, and 97.42% for the AC–DC rectifier, isolated DC–DC converter, and DC–AC inverter, respectively, resulting in an overall efficiency of 91.07%. Harmonic analysis using the FFT tool in MATLAB/Simulink produced an output voltage THD of 4.44% and an output current THD of 1.61%, demonstrating good output waveform quality.

The numerical efficiency and harmonic results reported in this Report are simulation-based. The preliminary hardware prototype documents the practical implementation of the proposed topology, while complete calibrated experimental validation remains future work. The developed 50 W Solid-State Transformer demonstrates effective electrical isolation and satisfactory simulated harmonic performance, providing a foundation for future research in advanced power electronic transformer technologies.

Keywords: Solid-State Transformer, High-Frequency Transformer, SPWM, MATLAB/Simulink, Arduino Mega 2560, Power Electronics, Total Harmonic Distortion, Power Quality, Efficiency.

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Abbreviations

AC	Alternating current
ADC	Analogue-to-digital converter
DAB	Dual-active bridge
DC	Direct current
EMI	Electromagnetic interference
ESR	Equivalent series resistance
FFT	Fast Fourier transform
GaN	Gallium nitride
HFT	High-frequency transformer
IGBT	Insulated-gate bipolar transistor
LC	Inductor-capacitor network
MOSFET	Metal-oxide-semiconductor field-effect transistor
PWM	Pulse-width modulation
RL	Resistor-inductor load
RMS	Root mean square
SiC	Silicon carbide
SPWM	Sinusoidal pulse-width modulation
SST	Solid-state transformer
THD	Total harmonic distortion
WBG	Wide-bandgap semiconductor

Symbols

V_{DC1}	Average voltage of the primary DC link
I_{DC1}	Average current of the primary DC link
V_{DC2}	Average voltage of the isolated DC link
I_{DC2}	Average current of the isolated DC link
f_s	Switching frequency
f_o	Output fundamental frequency
m_a	Amplitude modulation index
t_d	Dead time
L_f	Output-filter inductance
C_f	Output-filter capacitance
R_{on}	Effective MOSFET on-state resistance
η	Efficiency
ω	Angular frequency
ϕ	Phase angle

Chapter 1

Introduction

1.1 Background

Conventional transformers provide efficient voltage conversion and galvanic isolation, but their passive construction cannot independently regulate voltage, create an accessible DC port, control power flow or actively improve waveform quality. Modern networks increasingly contain converter-interfaced renewable sources, storage systems, electric-vehicle chargers and nonlinear loads. These applications motivate the solid-state transformer (SST), which combines semiconductor converters, a high-frequency transformer (HFT), digital control and measurement in one conversion chain [1, 2, 3].

The present work develops a safe, low-voltage, single-phase SST for simulation and laboratory study. A 15.5 V RMS source feeds a full-bridge rectifier and primary DC link. An IRF3205 MOSFET H-bridge, driven by IR2110 gate drivers, excites a 1:1 ferrite HFT at 25 kHz. The isolated DC link supplies an SPWM inverter followed by a 4.2 mH/20 μ F LC filter and a 4.7 Ω + 5 mH load. An Arduino Mega 2560 is used for the intended control implementation.

The frozen simulation values are $V_{DC1} = 20.4614$ V, $I_{DC1} = 1.4927$ A, $V_{DC2} = 20.2200$ V and $I_{DC2} = 1.4859$ A. The input and output powers are 32.1855 W and 29.3109 W. The rectifier, isolation and output-stage efficiencies are 95.0097%, 98.3887% and 97.4214%, giving 91.07% overall efficiency. Output-voltage and output-current THD are 4.44% and 1.61%, respectively.

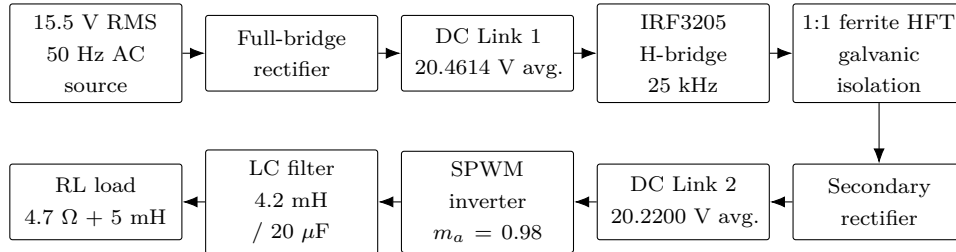


Figure 1.1: Functional architecture of the low-voltage single-phase SST.

1.2 Need for a Solid-State Transformer

A conventional transformer has a fixed turns ratio and its terminal voltage varies with source voltage, leakage impedance, current and power factor. Tap changers provide adjustment, but they are comparatively slow and do not offer cycle-by-cycle control. An SST places controlled conversion stages in the power path, allowing output regulation within the available DC-link and semiconductor limits.

A multi-stage SST also provides intermediate DC links for measurement, storage or DC loads. PWM and filtering can shape the reconstructed output rather than simply transferring source distortion. For a periodic output voltage, total harmonic distortion is

$$\text{THD}_v = \frac{\sqrt{\sum_{h=2}^{\infty} V_h^2}}{V_1} \times 100\%. \quad (1.1)$$

where V_1 is the RMS fundamental and V_h is the RMS value of harmonic order h . The same definition applies to current. Digital control additionally enables monitoring and rapid shutdown, although semiconductor devices require faster protection than a conventional transformer because of their lower short-circuit energy tolerance [2, 4].

1.3 Motivation and Problem Statement

Utility-scale SST prototypes require specialised high-voltage devices, insulation and test facilities. A low-voltage model allows the same principles to be studied with commonly available components. The design must convert AC to isolated AC, avoid bridge shoot-through, maintain balanced HFT excitation, use the isolated DC voltage effectively, attenuate carrier-frequency ripple and keep cumulative losses consistent with the reported overall efficiency.

1.4 Aim and Objectives

The aim is to design, model and evaluate a low-voltage single-phase SST and identify practical measures for improving waveform quality and efficiency. The objectives are to:

1. review the SST topologies and principles relevant to the selected design;
2. model the rectifier, DC links, high-frequency bridge, 1:1 ferrite HFT, SPWM inverter, LC filter and RL load;

3. define Arduino Mega 2560 and IR2110 control requirements, including complementary switching and dead time;
4. evaluate voltages, currents, power flow, stage efficiencies and FFT-based THD using a consistent parameter set;
5. compare simulation and laboratory behaviour; and
6. identify limitations and realistic improvements for future hardware.

1.5 Scope

The work covers mathematical modelling, MATLAB/Simulink evaluation and a laboratory-scale hardware prototype using IRF3205 MOSFETs, IR2110 drivers, an Arduino Mega 2560 and a ferrite HFT. It does not claim medium-voltage insulation qualification, grid-code certification, EMI compliance or long-duration reliability testing. The reported forward drops and on-state resistance are simulation parameters, and the 1:1 HFT is used to demonstrate isolation rather than voltage scaling. The achieved output power is approximately 29.31 W under the selected source, load and modulation conditions, although the platform was conceived as a 50 W laboratory demonstrator.

1.6 Methodology

The work proceeds from literature review and specification freezing to stage-by-stage modelling, control-signal definition, complete simulation, FFT analysis, hardware construction and performance comparison. Power and efficiency are evaluated over steady-state windows and checked for consistency across the three stages.

results.

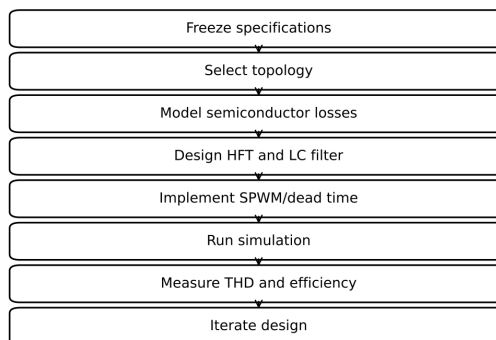


Figure 1.2: Methodology used for the dissertation and final design evaluation.

For cascaded stages, overall efficiency is the product of the individual stage efficiencies. Using the frozen results gives

$$\eta_{\text{overall}} = 0.950097 \times 0.983887 \times 0.974214 = 0.9107,$$

or 91.07%.

1.7 Significance and Dissertation Organisation

The study converts the broad SST concept into a reproducible low-voltage architecture and connects the final THD and efficiency values to practical component and control choices. Chapter 2 reviews the directly relevant SST fundamentals. Chapter 3 presents the detailed design and modelling. Chapter 4 describes control, simulation and performance enhancement. Chapter 5 reports the results, hardware prototype, conclusions and future scope. Appendices A and B retain the frozen engineering data and the essential control and hardware notes needed to reproduce the reported values safely.

Literature Review and Solid-State Transformer Fundamentals

2.1 SST Development and Classification

An SST is a power-electronic system that combines voltage conversion, galvanic isolation and controlled power flow. Research progressed from early high-frequency-link converters to smart-grid interfaces capable of integrating AC and DC systems, storage and distributed generation [1, 2, 3, 5]. The main architectures are classified by the number of conversion stages [6].

A single-stage SST directly converts input AC to a high-frequency waveform and reconstructs the output after the HFT. It uses fewer energy-storage components but couples input and output control and requires difficult commutation. Two-stage systems introduce one principal DC link, improving dynamic decoupling and providing a connection point for storage or DC loads. Three-stage systems normally use an active AC–DC rectifier, an isolated DC–DC converter and a DC–AC inverter. They offer the greatest control flexibility but have the highest device count, control burden and cumulative loss. High-voltage demonstrations using SiC devices show the potential of compact direct conversion, but specialised devices, magnetics and protection remain necessary [7].

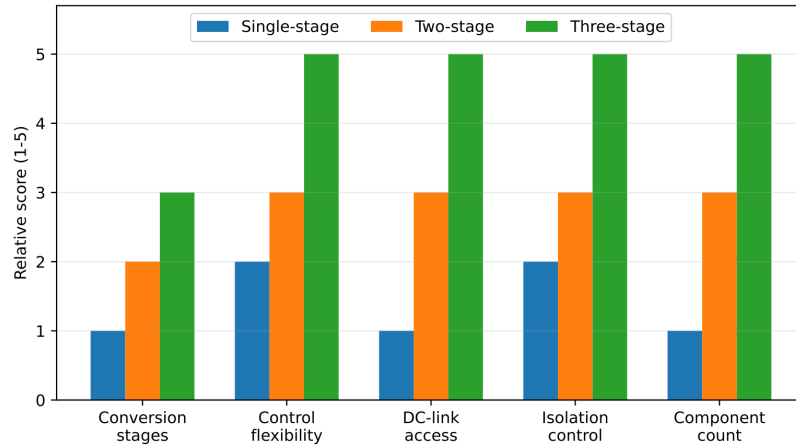


Figure 2.1: Qualitative comparison of SST topologies by conversion stage.

Table 2.1: Comparison of major SST topology families.

Characteristic	Single-stage	Two-stage	Three-stage
Intermediate DC link	Normally absent	One principal link	Two regulated links
Control decoupling	Limited	Moderate	High
Device count	Lowest	Moderate	Highest
Commutation burden	High for direct AC-AC forms	Moderate	Distributed among stages
Auxiliary DC port	Difficult	Available	Available at multiple levels
Typical strength	Compact conversion	Balance of control and efficiency	Maximum functionality
Typical limitation	Coupled control and protection	Reduced port flexibility	Cost, loss and complexity

The selected laboratory system follows the functional sequence of a three-stage SST, but with a passive input bridge and a unity-ratio isolation link. This simplification preserves electronic conversion, high-frequency isolation and controlled output reconstruction without the complexity of a utility-scale active front end.

2.2 Converter Building Blocks

2.2.1 Rectifier and DC Link

A diode bridge is inexpensive, rugged and suitable for a low-power demonstrator. Its limitations are non-sinusoidal input current, no regenerative operation and limited DC-link regulation. An active PWM rectifier can regulate the bus and shape input current, but requires additional switches, sensors and control. For a 15.5 V RMS source, the sinusoidal peak is 21.92 V. With two 0.2 V diode drops in the conducting path, the approximate no-load rectified peak is 21.52 V; loading and ripple reduce the final average to 20.4614 V.

2.2.2 High-Frequency Bridge and Isolation

A voltage-source H-bridge applies alternating positive and negative DC-link voltage to the HFT. Opposite diagonal pairs must switch symmetrically so that the average primary voltage is zero. Dead time prevents shoot-through, while excessive dead time increases body-diode conduction and waveform error. At 25 kHz, the switching period is 40 μs , allowing approximately 19 μs diagonal conduction intervals separated by 1 μs dead time.

The HFT provides galvanic isolation and permits a much smaller magnetic core than a 50 Hz transformer. For square-wave excitation, peak flux density is approximated by

$$B_{\text{pk}} = \frac{V_p}{4f_s N_p A_e}. \quad (2.1)$$

where V_p is primary voltage, N_p is primary turns and A_e is effective core area. Equation (2.1) explains the reduction in required turns at high frequency and the need for balanced volt-seconds. Ferrite is suitable at tens of kilohertz because of its high resistivity and low eddy-current loss. Copper loss, core loss, leakage inductance, insulation and temperature still limit practical power density [8, 9].

A dual-active bridge adds a controlled bridge on both sides of the HFT and regulates bidirectional power through phase shift [10]. The present system uses a simpler secondary conversion arrangement because bidirectional transfer is outside the project scope.

2.2.3 Gate Drivers and Digital Control

The IR2110 provides high-side and low-side outputs for one bridge leg. Its floating high-side supply is maintained by a bootstrap diode and capacitor, so the switching pattern must periodically recharge the bootstrap capacitor. Local decoupling, short gate loops, gate resistors and pull-down resistors are necessary in hardware.

The Arduino Mega 2560 offers enough timers and digital outputs for the bridge and SPWM signals. Hardware timers are preferable to software delays because deterministic timing is required for complementary switching and dead time. It is suitable for the low-voltage prototype but is not proposed as an industrial controller for a medium-voltage SST.

2.3 Output Inverter, SPWM and Filter

The output bridge synthesises a 50 Hz waveform by comparing a sinusoidal reference with a high-frequency carrier. The selected amplitude modulation index, $m_a = 0.98$,

uses most of the isolated DC bus while retaining a margin below overmodulation. Bipolar SPWM switches the bridge between positive and negative bus voltage; unipolar SPWM introduces a zero level and can reduce ripple for the same device switching frequency. More advanced methods such as selective harmonic elimination, space-vector PWM and multilevel conversion are important at higher power but are unnecessary for this single-phase prototype [11].

The LC filter separates the 50 Hz fundamental from the 25 kHz switching components. Its undamped resonance is

$$f_0 = \frac{1}{2\pi\sqrt{L_f C_f}}. \quad (2.2)$$

Using $L_f = 4.2$ mH and $C_f = 20$ μ F gives approximately 549 Hz, satisfying $50 \text{ Hz} \ll f_0 \ll 25 \text{ kHz}$. The $4.7 \text{ } \Omega + 5 \text{ mH}$ load provides damping and produces a lagging current. A lower cutoff frequency would improve switching-ripple attenuation but increase fundamental voltage drop and slow the response; therefore filter values must balance THD, regulation and stability.

2.4 Semiconductor Choice and Loss Mechanisms

IRF3205 silicon MOSFETs are appropriate for the approximately 20 V bus because they combine low on-state resistance with adequate current capability. Their gate charge and temperature-dependent resistance remain important at 25 kHz. SiC MOSFETs offer higher voltage capability and lower switching loss for medium-voltage modular SSTs, while GaN devices are attractive for compact lower-voltage high-frequency converters. Wide-bandgap devices can improve power density but demand careful gate-drive design, layout and EMI control.

SST losses arise from rectifier drops, MOSFET conduction and switching, gate-drive power, HFT copper and core loss, DC-link capacitor ESR, filter resistance and reactive-current circulation. Because overall efficiency is the product of the stage efficiencies, apparently small losses at each stage accumulate. This is a central disadvantage relative to a conventional transformer operating near rated load.

2.5 Applications and Implementation Challenges

SSTs are studied for smart distribution networks, renewable-energy interfaces, battery systems, electric-vehicle fast charging, traction and aerospace applications. Their controllable AC and DC ports can regulate voltage, exchange reactive power and coordinate storage. The benefits are strongest where these functions replace separate

converters or where reduced size has high value.

The same switching action creates practical challenges. Fast voltage and current transitions cause conducted and radiated EMI and can transfer common-mode current through HFT parasitic capacitance. Semiconductor devices have limited fault-energy tolerance, requiring rapid overcurrent detection and reliable gate shutdown. Concentrated semiconductor and magnetic losses demand thermal design. Driver supplies, sensors, firmware, cooling and DC-link capacitors also become part of the reliability chain [2, 4]. These issues explain why SST adoption depends on system-level benefits rather than power density alone.

2.6 Research Gap and Selected Architecture

Many educational SST reports mix incompatible ratings, omit gate-drive details, report THD without the FFT window or filter values, and quote efficiency without a complete power balance. The present work addresses these weaknesses by freezing one parameter set, stating the non-ideal switch values, reporting both DC links, using stage-wise efficiencies and linking the final THD to a defined load and filter.

The selected architecture prioritises reproducibility. The diode bridge avoids an additional controlled input stage; IRF3205 devices suit the low DC voltage; IR2110 drivers provide a practical half-bridge interface; the 1:1 ferrite HFT isolates without adding a voltage-ratio variable; and the Arduino Mega provides accessible timer resources. The 25 kHz switching frequency balances magnetic size, switching loss and implementation difficulty. The following chapter converts these choices into the detailed mathematical and simulation model.

Design and Modelling of the Proposed Single-Phase SST

3.1 Introduction

This chapter develops the proposed SST from its frozen specification to a stage-by-stage mathematical model. The system is divided into an input rectifier and DC link, a high-frequency MOSFET bridge, a 1:1 ferrite-core transformer, a secondary DC link, an SPWM output bridge, an LC filter and a series RL load. The same parameter set is used in all equations, tables and performance calculations. The purpose of the model is twofold. First, it establishes that the selected values are electrically consistent. Second, it provides an implementation reference for a low-voltage prototype. The model includes non-ideal switch and diode parameters so that efficiency is not evaluated with ideal semiconductor blocks.

3.2 Frozen Design Specification

Table 3.1 lists the values that define the final model. These values supersede the ratings of the earlier three-phase utility-scale report.

Table 3.1: Frozen specification of the proposed single-phase SST.

Parameter	Final value	Engineering role
Input supply	15.5 V RMS, 50 Hz	Low-voltage isolated source
Input rectifier	Single-phase full bridge	AC-to-DC conversion
Controller	Arduino Mega 2560	Gate timing and SPWM generation
Power MOSFET	IRF3205	Low-voltage bridge switching
Gate driver	IR2110	High-side and low-side drive
High-frequency transformer	Ferrite core, 1:1	Galvanic isolation at 25 kHz
Switching frequency	25 kHz	HFT and PWM carrier frequency
Switching period	40 μ s	Reciprocal of switching frequency
Modulation index	0.98	Fundamental output control
Output filter	4.2 mH and 20 μ F	Switching-ripple attenuation
Load	4.7 Ω + 5 mH series RL	Representative inductive load
MOSFET on-state parameter	0.02 Ω	Conduction-loss model
Rectifier diode drop	0.2 V	Frozen simulation parameter
MOSFET body-diode drop	0.2 V	Frozen simulation parameter
V_{DC1} average	20.4614 V	Primary DC-link result
I_{DC1} average	1.4927 A	Primary DC-link result
V_{DC2} average	20.2200 V	Isolated DC-link result
I_{DC2} average	1.4859 A	Isolated DC-link result

3.3 System Decomposition and Power Flow

The complete system is represented by the cascade

$$\begin{aligned} \text{AC source} \rightarrow \text{rectifier} \rightarrow \text{DC link 1} \rightarrow \text{HF bridge} \rightarrow \text{HFT} \rightarrow \text{DC link 2} \\ \rightarrow \text{SPWM bridge} \rightarrow \text{LC filter} \rightarrow \text{RL load.} \end{aligned} \quad (3.1)$$

The isolation barrier lies inside the high-frequency link. Control and measurement circuits on opposite sides should not share an unintended conductive ground path in physical hardware. The input power is 32.1855 W. Applying the stage efficiencies gives the internal power sequence

$$P_1 = P_{\text{in}}\eta_1 = (32.1855)(0.950097) = 30.57935 \text{ W.} \quad (3.2)$$

$$P_2 = P_1\eta_2 = (30.57935)(0.983887) = 30.08662 \text{ W.} \quad (3.3)$$

$$P_{\text{out}} = P_2\eta_3 \approx (30.08662)(0.974214) = 29.3109 \text{ W.} \quad (3.4)$$

The stages therefore dissipate approximately 1.6062 W, 0.4927 W and 0.7757 W, respectively. The total modelled loss is approximately 2.8746 W.

3.4 Input Source Model

The source is represented by

$$v_s(t) = V_m \sin(\omega_o t), \quad (3.5)$$

where

$$V_m = \sqrt{2} \times 15.5 \text{ V} = 21.92 \text{ V}, \quad (3.6)$$

$$\omega_o = 2\pi \times 50 \text{ Hz} = 314.16 \text{ rad/s}. \quad (3.7)$$

The source is assumed to be isolated from the mains for prototype work. A bench isolation transformer or a dedicated low-voltage AC source is required. Direct connection of the low-voltage bridge to the utility supply is outside the scope and would be unsafe. The source impedance affects rectifier current peaks. An ideal zero-impedance source feeding a large capacitor produces unrealistically high charging current. In simulation, a small source resistance or transformer leakage impedance should be included. This also improves numerical stability and better represents a practical step-down transformer.

3.5 Full-Bridge Rectifier Design

3.5.1 Conduction States

The bridge rectifier uses two diodes during each half cycle. For $v_s(t) > 0$, one diagonal pair conducts; for $v_s(t) < 0$, the opposite pair conducts. The rectified voltage before filtering is approximately

$$v_r(t) = \max(|v_s(t)| - 2V_f, 0), \quad (3.8)$$

with $V_f = 0.2 \text{ V}$ in the frozen model.

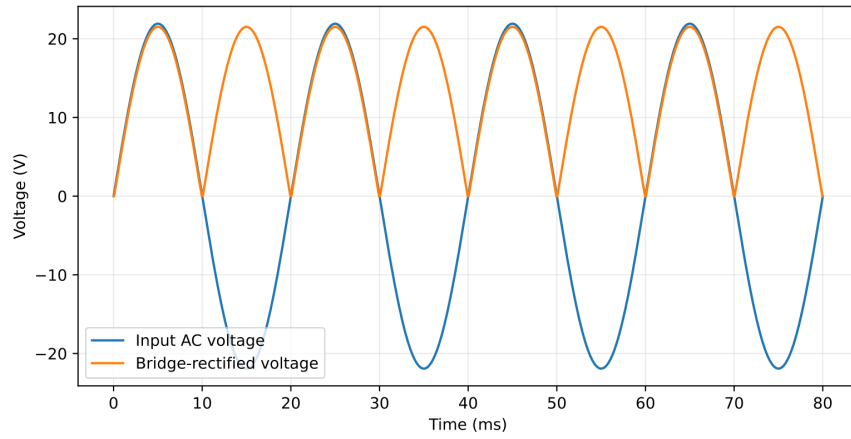


Figure 3.1: Input sinusoid and bridge-rectified voltage for the frozen input and diode-drop parameters.

The theoretical no-load peak is 21.52 V. The final average of 20.4614 V indicates a loaded DC bus with a moderate ripple and finite conduction loss. The rectifier output frequency is 100 Hz, twice the source frequency.

3.5.2 DC-Link Capacitor Model

Between charging pulses, the capacitor supplies the downstream bridge. If the load current is approximately constant over a ripple interval, the peak-to-peak ripple can

be estimated by

$$\Delta V_{\text{DC}} \approx \frac{I_{\text{DC}}}{2f_o C_{\text{DC}}}. \quad (3.9)$$

This equation can be rearranged for a desired ripple:

$$C_{\text{DC}} \geq \frac{I_{\text{DC}}}{2f_o \Delta V_{\text{DC}}}. \quad (3.10)$$

For example, limiting ripple to 1 V at 1.4927 A would require roughly

$$C_{\text{DC}} \geq \frac{1.4927}{2 \times 50 \times 1} = 14.93 \text{ mF}. \quad (3.11)$$

This illustrates why capacitor-input rectifiers at low voltage and ampere-level current often require several millifarads. The exact capacitor value is not part of the frozen set, so it remains an implementation parameter. The simulation result, rather than

an assumed capacitor, governs the reported DC average.

The DC-link capacitor should be selected by ripple-current rating as well as capacitance. A physically large capacitance can reduce voltage ripple while increasing inrush current. A current-limited bench source or a pre-charge resistor is recommended during hardware commissioning.

3.5.3 Rectifier Loss Estimate

The average diode conduction loss can be approximated by

$$P_D \approx 2V_f I_{DC}, \quad (3.12)$$

which gives

$$P_D \approx 2 \times 0.2 \times 1.4927 = 0.5971 \text{ W}. \quad (3.13)$$

Additional loss appears in source resistance, capacitor ESR and current-pulse RMS heating. The frozen first-stage loss of approximately 1.6062 W is therefore plausible when all non-idealities are included.

3.6 Primary DC-Link Operating Point

The average primary DC-link power based on the measured average quantities is

$$P_{DC1,avg} = V_{DC1} I_{DC1} = (20.4614)(1.4927) = 30.5427 \text{ W}. \quad (3.14)$$

This is close to the stage-output power obtained from the efficiency chain. Small differences arise because the product of separately averaged voltage and current is not always identical to the average of instantaneous power:

$$\overline{v(t)} \overline{i(t)} \neq \overline{v(t)i(t)} \quad (3.15)$$

when ripple is correlated. Instantaneous power measurement is therefore preferred for precise efficiency calculation.

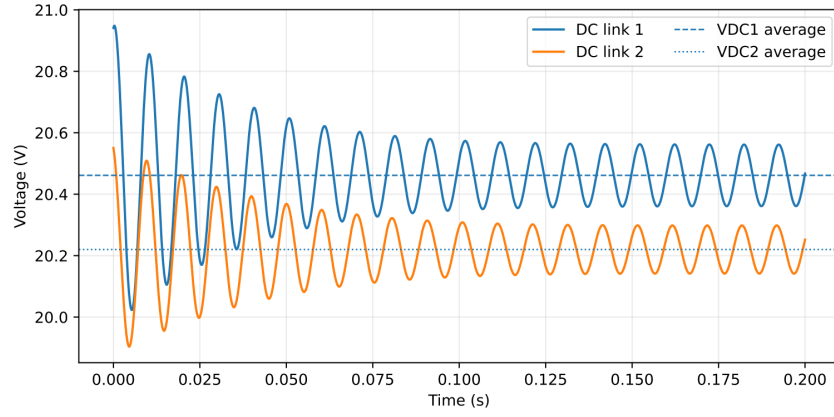


Figure 3.2: Representative DC-link settling and ripple around the frozen average values.

3.7 High-Frequency H-Bridge Design

3.7.1 Switch Arrangement

The bridge contains four IRF3205 MOSFETs. The intended diagonal sequence is

$$Q_1, Q_4 : +V_{DC1} \text{ across the HFT primary,} \quad (3.16)$$

$$Q_2, Q_3 : -V_{DC1} \text{ across the HFT primary.} \quad (3.17)$$

All switches are turned off briefly during each transition. This dead time prevents a high-side and low-side device in the same leg from conducting simultaneously.

3.7.2 Timing Calculation

At 25 kHz, the period is 40 μs . A practical sequence is

1. turn on Q_1 and Q_4 for approximately 19 μs ;
2. turn all devices off for approximately 1 μs ;
3. turn on Q_2 and Q_3 for approximately 19 μs ; and
4. turn all devices off for approximately 1 μs .

The corresponding duty ratio per polarity is about 0.475. The small reduction from

0.5 prevents overlap and gives the HFT a symmetrical bipolar excitation.

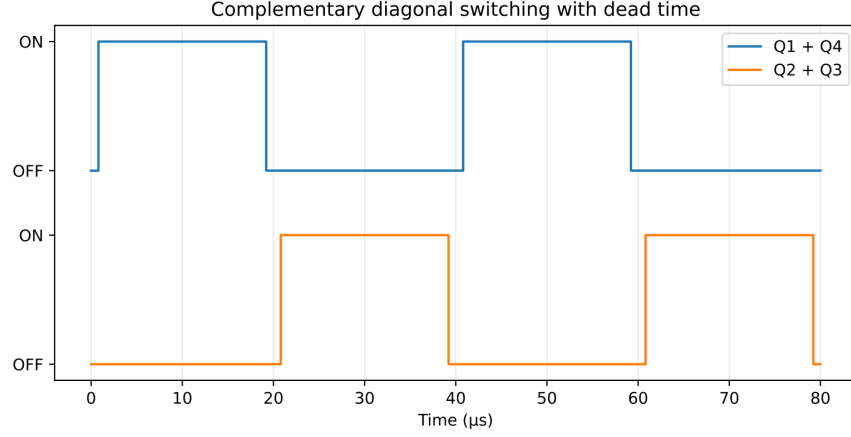


Figure 3.3: Complementary diagonal gate timing at 25 kHz.

3.7.3 MOSFET Conduction Loss

For a square-wave transformer current with RMS value $I_{p,rms}$, two MOSFETs conduct at a time. A first-order bridge loss estimate is

$$P_{bridge,cond} \approx 2I_{p,rms}^2 R_{on}. \quad (3.18)$$

If $I_{p,rms} \approx 1.5$ A and $R_{on} = 0.02$ Ω ,

$$P_{bridge,cond} \approx 2(1.5)^2(0.02) = 0.09 \text{ W}. \quad (3.19)$$

The actual loss is higher after switching loss, body-diode conduction, gate-drive loss and transformer magnetising current are included. The low conduction estimate confirms that the selected MOSFET has ample current capability at this operating point.

3.7.4 Switching Loss and Gate Drive

A simple transition-overlap estimate is

$$P_{sw} \approx \frac{1}{2} V_{DS} I_D (t_r + t_f) f_s N_{tr}, \quad (3.20)$$

where N_{tr} accounts for the number of relevant switching transitions. Gate-drive power is

$$P_g = N_g Q_g V_g f_s, \quad (3.21)$$

where N_g is the number of driven gates. These losses are not dominant at 20 V and 25 kHz, but poor gate resistance or slow drive can increase transition time and bridge heating. A $10\ \Omega$ gate resistor, as used in the intended hardware, limits peak gate current and damps ringing. The resistance may be adjusted after observing the gate waveform, drain overshoot and switching temperature. The gate-source path should also include a pull-down resistor so that the MOSFET remains off while the controller is resetting.

3.8 IR2110 Driver Interface

Each IR2110 drives one bridge leg. The key functional connections are:

- VCC to the regulated gate-drive supply;
- COM to the local power ground;
- LO to the low-side MOSFET gate through a gate resistor;
- VS to the half-bridge switch node;
- HO to the high-side MOSFET gate through a gate resistor;
- VB to the bootstrap capacitor and bootstrap diode;
- HIN and LIN to complementary controller logic; and
- local ceramic decoupling between VCC and COM.

The high-side output is referenced to VS rather than to system ground. Measuring HO incorrectly with a ground-referenced oscilloscope can short the switching node through the probe ground. Differential probing or isolated measurement is required. In low-voltage initial testing, the bridge should be powered from a current-limited supply and the gate waveforms should be verified before the HFT is connected.

3.9 High-Frequency Transformer Design

3.9.1 Turns Ratio and Voltage Transfer

The frozen ratio is

$$n = \frac{N_s}{N_p} = 1. \quad (3.22)$$

For an ideal transformer,

$$\frac{V_s}{V_p} = \frac{N_s}{N_p} = 1, \quad \frac{I_s}{I_p} = \frac{N_p}{N_s} = 1. \quad (3.23)$$

The measured isolated DC average 20.2200 V is slightly below 20.4614 V, consistent with bridge, transformer, rectifier and DC-link losses.

3.9.2 Volt-Second Balance

A ferrite transformer must not receive a net DC volt-second. For one switching period,

$$\int_0^{T_s} v_p(t) dt = 0. \quad (3.24)$$

Symmetrical positive and negative pulses satisfy this condition. Unequal pulse widths or unequal device drops can create flux walking. The controller should therefore generate both diagonal pulses from one timer base and use matched dead time.

3.9.3 Minimum Turns

Rearranging Equation (2.1), the minimum primary turns for a selected peak flux density is

$$N_p \geq \frac{V_p}{4f_s A_e B_{\max}}. \quad (3.25)$$

As an illustrative calculation, if $V_p = 20.5$ V, $A_e = 125 \text{ mm}^2$ and $B_{\max} = 0.18$ T,

$$N_p \geq \frac{20.5}{4(25,000)(125 \times 10^{-6})(0.18)} \approx 9.1. \quad (3.26)$$

A design would therefore use at least 10 turns, with margin for waveform asymmetry, temperature and core material. The numerical core area is illustrative; the final turns must be calculated from the actual core datasheet.

3.9.4 Magnetising Current

The magnetising inductance L_m causes a triangular current under square-wave excitation. During one positive interval,

$$\Delta I_m = \frac{V_p \Delta t}{L_m}. \quad (3.27)$$

If L_m is too small, magnetising current increases conduction loss without delivering useful load power. Measuring the inductance of the completed primary with the secondary open is therefore an important prototype check.

3.9.5 Leakage Inductance and Spikes

Leakage inductance stores energy that cannot instantaneously transfer between windings. When the bridge commutates, this energy can produce drain-voltage overshoot. Tight primary-secondary coupling reduces leakage but increases inter-winding capacitance and common-mode current. Winding arrangement is therefore a trade-off. RC or RCD snubbers may be required after measuring switching spikes.

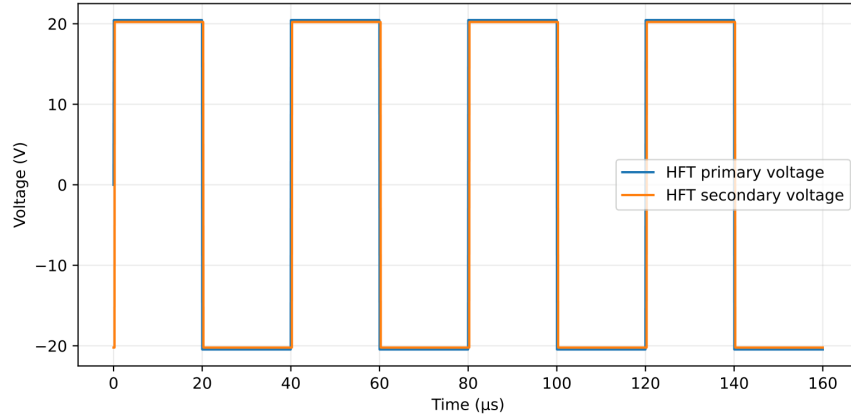


Figure 3.4: Representative primary and secondary square-wave voltages for the 1:1 high-frequency link.

3.10 Secondary Rectification and DC Link

The HFT secondary waveform is rectified and filtered before the output inverter. With a unity ratio, the secondary DC level is expected to be close to the primary level. The final values show

$$\Delta V_{\text{isolation}} = V_{\text{DC1}} - V_{\text{DC2}} = 0.2414 \text{ V.} \quad (3.28)$$

The corresponding relative reduction is

$$\frac{0.2414}{20.4614} \times 100\% \approx 1.18\%. \quad (3.29)$$

This small voltage difference is consistent with the high efficiency 98.3887 % of the high-frequency isolation stage. The average isolated-side DC power from averaged quantities is

$$V_{\text{DC2}} I_{\text{DC2}} = 20.2200 \times 1.4859 \approx 30.045 \text{ W.} \quad (3.30)$$

This agrees closely with the stage-output value 30.0866 W. Again, the remaining difference is attributable to averaging and waveform ripple.

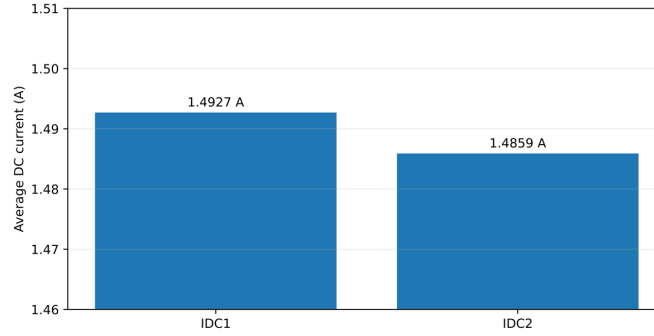


Figure 3.5: Comparison of the frozen primary and isolated DC-current averages.

3.11 Output SPWM Inverter

3.11.1 Reference and Carrier Signals

The output stage uses a 50 Hz sinusoidal reference and a 25 kHz triangular carrier. The reference is

$$v_{\text{ref}}(t) = m_a \sin(2\pi f_o t), \quad (3.31)$$

with $m_a = 0.98$. The comparator output determines the switching state. Complementary gate signals are generated for each bridge leg with dead time.

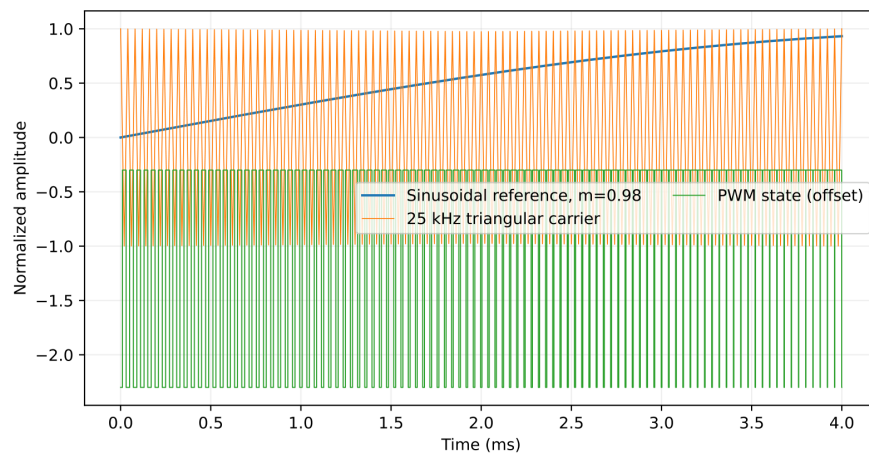


Figure 3.6: SPWM reference, carrier and representative switching state.

3.11.2 Fundamental Voltage Estimate

For a full-bridge bipolar SPWM inverter operating in the linear region, the fundamental output peak can be approximated by

$$V_{1,\text{pk}} \approx m_a V_{\text{DC}2}. \quad (3.32)$$

The corresponding RMS value is

$$V_{1,\text{rms}} \approx \frac{m_a V_{\text{DC}2}}{\sqrt{2}} = \frac{0.98 \times 20.22}{\sqrt{2}} \approx 14.01 \text{ V}. \quad (3.33)$$

Filter drop and losses reduce the actual load voltage. This is an ideal pre-filter estimate. From the reported active output power and the specified $4.7 \Omega + 5 \text{ mH}$ load, the load-side RMS voltage is approximately 12.38 V; the difference is attributable to filter drop, switching effects and model losses.

3.11.3 Modulation Margin

At $m_a = 0.98$, the design is close to the edge of linear modulation. A small margin remains for DC-link ripple and controller quantisation. Increasing m_a above unity would enter overmodulation, potentially increasing the fundamental voltage but also increasing low-order distortion. The chosen value therefore favours maximum utilisation without deliberately operating in a nonlinear modulation region.

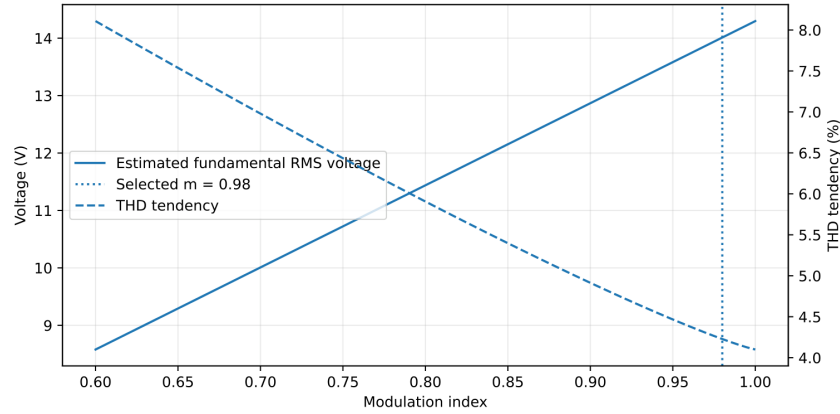


Figure 3.7: Qualitative sensitivity of fundamental voltage and distortion to modulation index.

3.12 LC Output Filter Design

3.12.1 Cutoff and Resonance

The filter values are

$$L_f = 4.2 \text{ mH}, \quad C_f = 20 \text{ }\mu\text{F}. \quad (3.34)$$

Equation (2.2) gives approximately 549 Hz. The frequency ratio relative to the output fundamental is

$$\frac{f_0}{f_o} \approx \frac{549}{50} \approx 10.98, \quad (3.35)$$

while the ratio relative to the carrier is

$$\frac{f_s}{f_0} \approx \frac{25,000}{549} \approx 45.5. \quad (3.36)$$

This placement provides a broad separation between the 50 Hz signal and the switching band.

3.12.2 Inductor Fundamental Reactance

At 50 Hz, the filter-inductor reactance is

$$X_{L_f} = 2\pi f_o L_f = 2\pi(50)(4.2 \times 10^{-3}) \approx 1.319 \Omega. \quad (3.37)$$

This is not negligible compared with the load resistance. The output controller or modulation index must compensate for the associated voltage drop. At 25 kHz,

$$X_{L_{f,s}} = 2\pi(25,000)(4.2 \times 10^{-3}) \approx 659.7 \Omega, \quad (3.38)$$

so the inductor strongly opposes switching ripple.

3.12.3 Capacitor Reactance

At 50 Hz,

$$X_{C_f} = \frac{1}{2\pi f_o C_f} \approx 159.2 \Omega, \quad (3.39)$$

which means the capacitor draws relatively little fundamental current. At 25 kHz,

$$X_{C_{f,s}} \approx 0.318 \Omega, \quad (3.40)$$

providing a low-impedance path for switching-frequency components. The large separation of reactances explains the strong filtering action.

3.12.4 Filter Transfer Behaviour

An ideal unloaded LC low-pass transfer function can be written as

$$G(s) = \frac{1}{L_f C_f s^2 + 1}. \quad (3.41)$$

With damping and load included, the denominator gains a first-order term. The actual poles depend on where the load is connected and on parasitic resistance. A lightly damped resonance can amplify components near 549 Hz; therefore, time-domain simulation should check start-up and load-step behaviour rather than relying only on steady-state FFT.

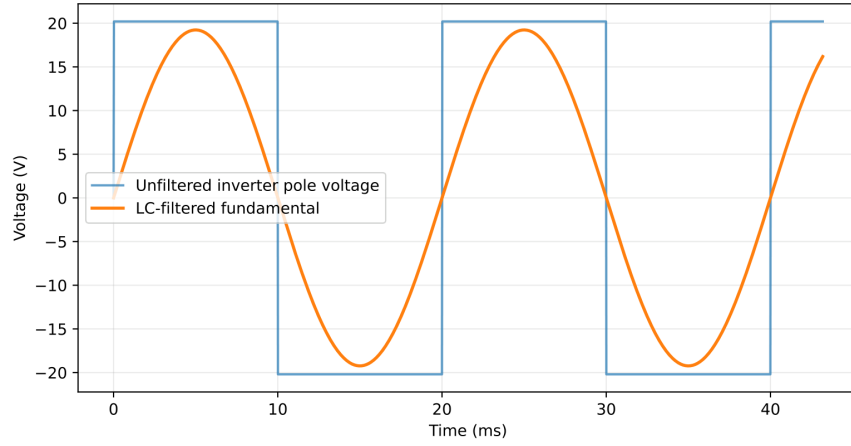


Figure 3.8: Conceptual effect of the LC filter on the inverter switching waveform.

3.13 RL Load Model

The load is a series combination

$$R_L = 4.7 \Omega, \quad L_L = 5 \text{ mH}. \quad (3.42)$$

At 50 Hz, the load inductive reactance is

$$X_{L_L} = 2\pi(50)(5 \times 10^{-3}) = 1.571 \Omega. \quad (3.43)$$

The magnitude of the load impedance is

$$|Z_L| = \sqrt{4.7^2 + 1.571^2} \approx 4.956 \Omega. \quad (3.44)$$

The power-factor angle is

$$\phi = \tan^{-1} \left(\frac{1.571}{4.7} \right) \approx 18.49^\circ, \quad (3.45)$$

and the displacement power factor is

$$\cos \phi \approx 0.948. \quad (3.46)$$

If the real output power is 29.3109 W, the approximate RMS load current

from $P = I^2 R$ is

$$I_{o,\text{rms}} \approx \sqrt{\frac{29.3109}{4.7}} \approx 2.498 \text{ A}. \quad (3.47)$$

The corresponding load-voltage magnitude is

$$V_{o,\text{rms}} \approx I_{o,\text{rms}} |Z_L| \approx 12.38 \text{ V}. \quad (3.48)$$

This back-calculation is compatible with a 20.22 V DC link and a modulation index of 0.98 after filter and converter drops. The load current obeys

$$L_L \frac{di_o}{dt} + R_L i_o = v_o(t). \quad (3.49)$$

For a sinusoidal steady state, current lags voltage by ϕ . The inductance also smooths high-frequency current ripple, helping to explain why the final current THD is lower than the voltage THD.

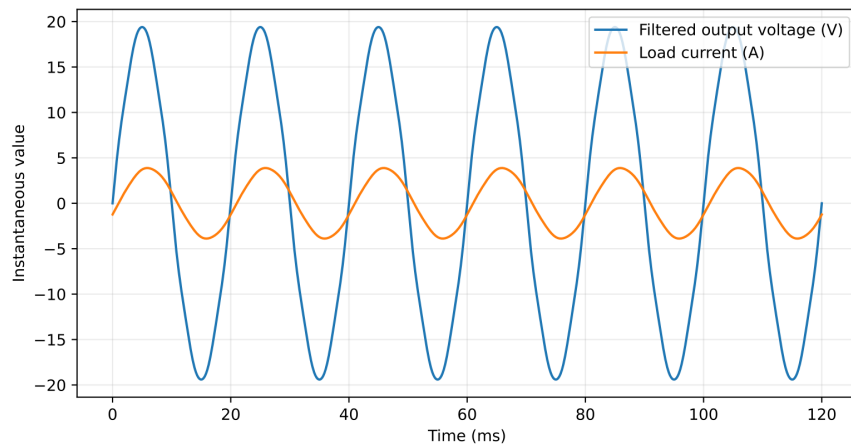


Figure 3.9: Representative filtered voltage and lagging RL-load current.

3.14 Small-Signal and State-Space Representation

For filter and load analysis, choose states

$$\mathbf{x} = \begin{bmatrix} i_f & v_C & i_L \end{bmatrix}^T, \quad (3.50)$$

where i_f is filter-inductor current, v_C is capacitor voltage and i_L is load current. A simplified averaged model is

$$L_f \frac{di_f}{dt} = v_{\text{inv}} - v_C - r_f i_f, \quad (3.51)$$

$$C_f \frac{dv_C}{dt} = i_f - i_L, \quad (3.52)$$

$$L_L \frac{di_L}{dt} = v_C - R_L i_L. \quad (3.53)$$

In matrix form,

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}v_{\text{inv}}, \quad (3.54)$$

with

$$\mathbf{A} = \begin{bmatrix} -r_f/L_f & -1/L_f & 0 \\ 1/C_f & 0 & -1/C_f \\ 0 & 1/L_L & -R_L/L_L \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1/L_f \\ 0 \\ 0 \end{bmatrix}. \quad (3.55)$$

This model is useful for designing a closed-loop voltage controller or active damping. The switching model remains necessary for THD and semiconductor-loss analysis.

3.15 Measurement Model

Voltage and current are measured at four principal locations:

1. source voltage and current;
2. primary DC-link voltage and current;
3. isolated DC-link voltage and current; and
4. filtered output voltage and load current.

Instantaneous power is computed as $p(t) = v(t)i(t)$ and averaged over an integer number of fundamental cycles. FFT analysis should use a steady-state window after start-up transients have decayed. The sample rate must be high enough to represent

25 kHz switching components. Averaging blocks can introduce misleading results if their window is not aligned with the fundamental. For a 50 Hz output, a window of 20 ms or an integer multiple is appropriate. THD should exclude the DC component and identify the fundamental correctly.

3.16 Loss Model

The total loss is decomposed as

$$P_{\text{loss}} = P_{\text{rect}} + P_{\text{MOS,cond}} + P_{\text{MOS,sw}} + P_{\text{gate}} + P_{\text{HFT,cu}} + P_{\text{HFT,core}} + P_{\text{filter}} + P_{\text{aux}}. \quad (3.56)$$

The frozen stage efficiencies implicitly include aggregated losses. The model does not attempt to identify every physical loss from device datasheets; instead, it checks that the total power balance is consistent. For hardware design, each term should be calculated at maximum temperature and current.

3.17 Design Consistency Checks

Several checks are performed before accepting the parameter set:

1. The rectified peak exceeds the primary DC average.
2. The isolated DC average is slightly below the primary DC average for a unity-ratio HFT.
3. The filter resonance is above the output fundamental and below the carrier.
4. The modulation index remains within the linear SPWM region.
5. The RL-load current is compatible with the reported output power.
6. The product of stage efficiencies reproduces the overall efficiency.
7. The output current THD is plausibly lower than voltage THD because of load inductance.

All checks are satisfied within the precision expected from simulation averages.

3.18 Chapter Summary

This chapter translated the frozen specification into a complete electrical model. The 15.5 V input produces a primary DC average of 20.4614 V, which feeds a 25 kHz MOSFET bridge and 1:1 ferrite transformer. The isolated DC link supplies an SPWM

bridge with $m_a = 0.98$, followed by a 4.2 mH/20 μ F filter and a 4.7 Ω + 5 mH series RL load. The equations establish expected voltage, current, phase angle, filter resonance and loss relationships. Chapter 4 now explains how the converter is controlled and how performance is enhanced without changing the frozen final result set.

Control Strategy, Simulation Model and Performance Enhancement

4.1 Introduction

The proposed SST contains two actively switched bridges and a high-frequency isolation stage. Its performance depends not only on component values but also on gate timing, modulation, measurement, start-up sequencing and numerical simulation settings. This chapter describes the controller structure, the implementation-oriented gate logic, the simulation workflow and the measures used to improve waveform quality and efficiency. The final reported values remain fixed. Performance enhancement therefore means explaining the design choices that led to the frozen result set and identifying further improvements that could be implemented without misrepresenting the achieved results.

4.2 Control Objectives

The primary control objectives are:

1. generate a symmetrical 25 kHz bipolar voltage for the HFT;
2. maintain dead time between complementary devices;
3. avoid a DC component in the transformer excitation;
4. generate a 50 Hz sinusoidal reference with $m_a = 0.98$;
5. reconstruct a low-distortion output after the LC filter;
6. provide a deterministic all-gates-off state during reset and fault conditions; and
7. measure power and THD after the model reaches steady state.

The frozen model is primarily open-loop. The DC-link voltages are not regulated by active front-end or DAB controllers. The output amplitude is set by the modulation index. This architecture simplifies implementation but makes the output sensitive to input-voltage and load variations.

4.3 Controller Hardware

4.3.1 Arduino Mega 2560 Resources

The Arduino Mega 2560 is based on the ATmega2560 microcontroller. It offers multiple 8-bit and 16-bit timers and enough I/O pins to generate separate logic signals for the bridge legs. Timer-generated outputs are preferred over software delays because they reduce jitter and allow complementary timing to be repeated accurately. A suitable logical allocation is:

- four digital outputs for the high-frequency H-bridge;
- four PWM or timer outputs for the output full bridge;
- one enable input for emergency shutdown;
- analog inputs for scaled voltage and current feedback; and
- status outputs for driver-ready and fault indication.

The exact pin mapping remains editable in the source and should match the assembled PCB or perfboard.

4.3.2 Clock and Timing Resolution

With a 16 MHz controller clock, one clock cycle is 62.5 ns. A 40 μ s switching period contains 640 clock cycles. A 1 μ s dead time corresponds to 16 clock cycles, which is readily representable. This resolution is adequate for the frozen switching frequency. If a timer uses a prescaler N , the compare value is

$$N_{\text{count}} = \frac{f_{\text{clk}}}{N f_s} - 1. \quad (4.1)$$

For $N = 1$ and $f_s = 25$ kHz, the period count is 639. The firmware can divide the period into positive-pulse, dead-time, negative-pulse and dead-time intervals.

4.4 High-Frequency Bridge Control

4.4.1 Diagonal Switching Sequence

The high-frequency bridge uses two diagonal states. A safe conceptual state table is shown in Table 4.1.

Table 4.1: High-frequency bridge switching states.

State	Q_1	Q_2	Q_3	Q_4	Primary voltage
Positive	1	0	0	1	$+V_{DC1}$
Dead time	0	0	0	0	Current-dependent
Negative	0	1	1	0	$-V_{DC1}$
Dead time	0	0	0	0	Current-dependent

The controller should update both legs from the same interrupt or timer event. Updating gates sequentially in ordinary program code can create brief unintended states. The outputs should default low when the microcontroller is unpowered or resetting.

4.4.2 Dead-Time Effects

Dead time prevents shoot-through but changes the average bridge voltage. During dead time, transformer current commutates through body diodes or output capacitances. The resulting voltage depends on current direction. At low output frequency,

dead-time distortion is often concentrated around current zero crossings. In the high-frequency transformer bridge, dead time reduces the effective volt-seconds and can cause asymmetry if the two transitions differ. A useful dead-time ratio is

$$d_d = \frac{t_d}{T_s} = \frac{1}{40} = 0.025. \quad (4.2)$$

Thus, approximately 2.5 % of each switching period is allocated to each $1 \mu\text{s}$ interval collectively across the two transitions. Reducing this interval can improve voltage utilisation, but only after verifying the real turn-off delay and gate waveform.

4.5 SPWM Control

4.5.1 Lookup-Table Method

A practical microcontroller implementation stores one sinusoidal cycle in a lookup table. A phase accumulator advances through the table at the desired output frequency. Each sample sets the PWM compare value. With carrier frequency f_s and output

frequency f_o , there are 500 switching periods per output cycle. A 500-sample table therefore maps naturally to one sample per carrier period, although a smaller table with interpolation or a phase accumulator can also be used. The duty command for a unipolar leg can be expressed as

$$d_a(k) = \frac{1}{2} [1 + m_a \sin \theta_k], \quad (4.3)$$

while the opposite leg uses

$$d_b(k) = \frac{1}{2} [1 - m_a \sin \theta_k]. \quad (4.4)$$

The differential fundamental is proportional to $d_a - d_b$.

4.5.2 Carrier Synchronisation

Synchronising the high-frequency link and output carrier can simplify measurement and reduce beat frequencies. However, the two bridges may also be operated from separate timers if their rates are identical and stable. A common clock ensures that phase drift does not create low-frequency modulation of the DC link.

4.5.3 Quantisation

Finite timer resolution quantises duty cycle. With 640 counts per period, the ideal duty resolution is about 0.156 %. This is adequate for the frozen modulation index. The practical resolution is lower if dead time and reserved counts reduce the usable range.

4.6 Start-Up and Shutdown Sequence

A controlled start-up reduces inrush, transformer flux imbalance and output transients. A recommended sequence is:

1. hold all gate outputs low;
2. energise the low-voltage AC source through current limiting;
3. allow the primary DC link to charge;
4. verify driver supply and fault inputs;
5. start the high-frequency bridge with equal positive and negative pulses;

6. confirm the isolated DC-link voltage;
7. ramp the output modulation index from zero to 0.98; and
8. enable normal monitoring.

Shutdown should first force the modulation command to zero, then disable both bridges. In a fault, all gates should be disabled immediately rather than waiting for a soft ramp.

4.7 Simulation Model Structure

The complete switching model contains the source, rectifier, primary capacitor, high-frequency bridge, HFT, secondary rectifier, isolated capacitor, output bridge, LC filter, RL load and measurement blocks. Semiconductor blocks use the frozen R_{on} and diode-drop parameters. The HFT is represented with magnetising and leakage inductance or a coupled-inductor block. The output PWM is generated from a sinusoidal reference and triangular carrier.

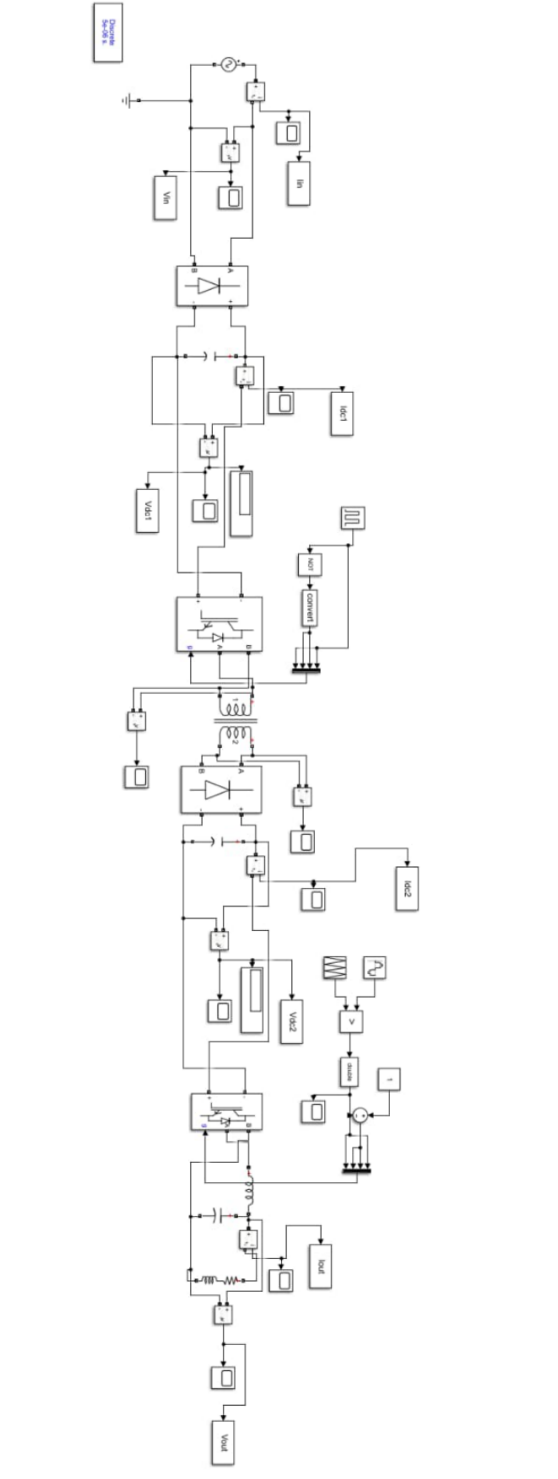


Figure 4.1: Simulink Model of the complete single-phase SST model.

4.7.1 Stage-wise Simulink Realisation

To make the model structure explicit, the uploaded stage-level Simulink screenshots are reproduced here. These figures show the original block-level implementation used to obtain the frozen results.

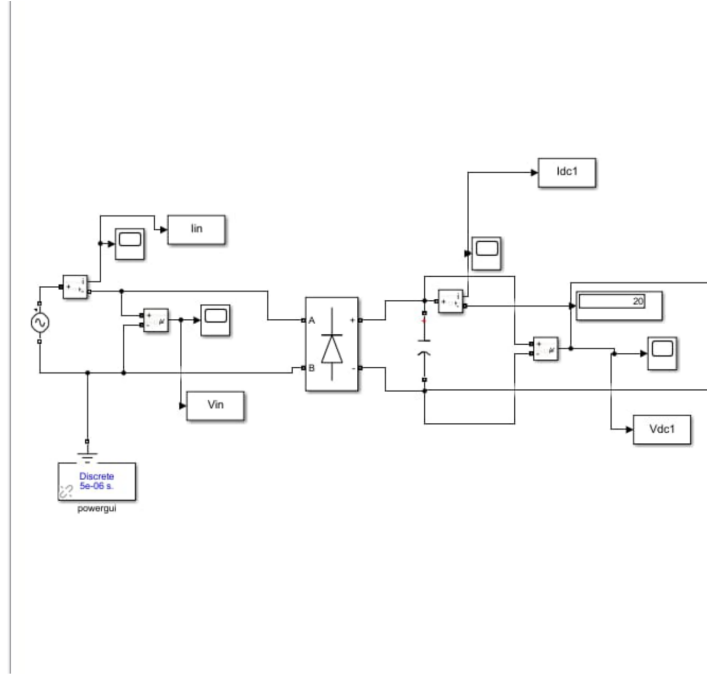


Figure 4.2: Simulink model of Stage 1: input AC source, rectifier and primary DC link.

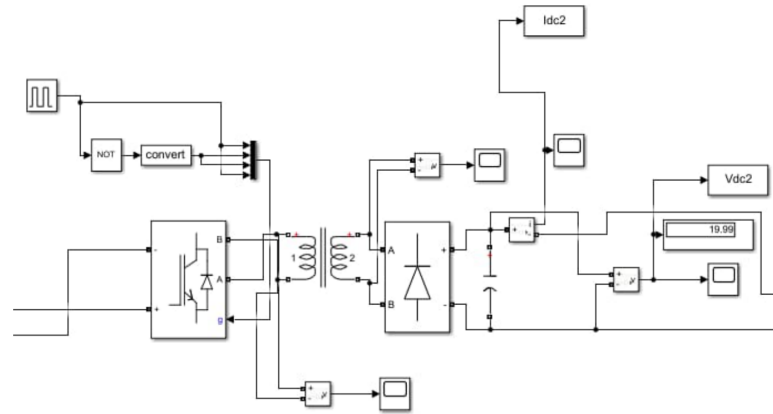


Figure 4.3: Simulink model of Stage 2: high-frequency bridge, ferrite transformer, secondary rectification and isolated DC link.

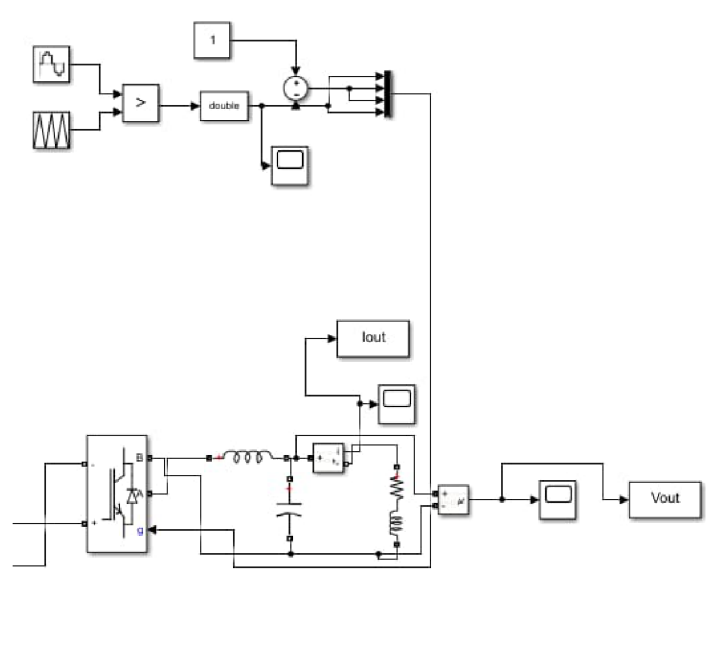


Figure 4.4: Simulink model of Stage 3: SPWM output bridge, LC filter and RL load.

4.7.2 Solver Selection

A switching model at 25 kHz contains fast discontinuities and a 50 Hz envelope. The maximum time step must be much smaller than the 40 μs switching period. A fixed step between 0.1 μs and 1 μs , depending on model stiffness and computer resources, can resolve switching edges. An averaged model can use a larger step but cannot provide switching THD directly. Stiff solvers can be useful when capacitors, inductors and ideal switches create widely separated time constants. The solver should be validated by reducing the step and confirming that average power and THD do not change significantly.

4.7.3 Initial Conditions

Starting both DC capacitors from zero can produce a long transient and large current pulse. Initial conditions may be set near the expected DC levels for steady-state analysis, but a separate zero-initial-condition run should be used to check start-up stress. The FFT window must begin after the transient has decayed.

4.7.4 Measurement Windows

For 50 Hz, a 100 ms window contains five cycles. A five- or ten-cycle steady-state window gives stable RMS and THD estimates. The window should contain an integer number of cycles to reduce spectral leakage. A Hann window may be used when exact coherence is unavailable, but it changes amplitude calibration and must be handled consistently.

4.8 Harmonic Analysis

4.8.1 Sources of Distortion

The output waveform contains distortion from several mechanisms:

- finite carrier frequency and PWM sidebands;
- dead time and body-diode conduction;
- DC-link ripple at 100 Hz;
- nonlinear switch voltage drops;
- HFT leakage and magnetising current;
- LC-filter resonance and damping; and

- numerical sampling and FFT leakage.

The final voltage THD of 4.44 % is therefore a system result, not solely a modulation result.

4.8.2 FFT Procedure

Let $x[n]$ be a sampled steady-state waveform. After removing any DC component, the discrete Fourier transform is

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}. \quad (4.5)$$

The fundamental bin corresponds to 50 Hz. Harmonic RMS components are extracted at integer multiples. THD is computed using Equation (1.1). The analysis bandwidth should be stated. Including switching-frequency components can produce a higher THD than considering only lower-order harmonics, so consistent settings are essential.

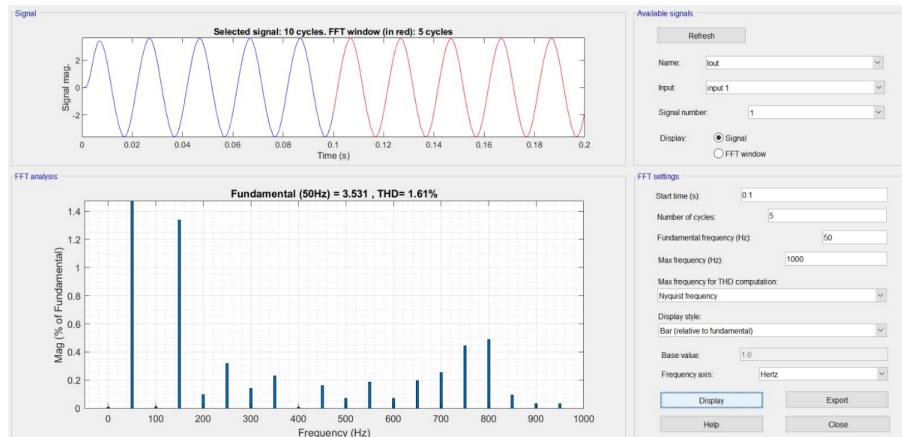


Figure 4.5: FFT-analysis window used in Simulink for the output-current harmonic evaluation. The analysis is carried out over a steady-state window after the transient has decayed.

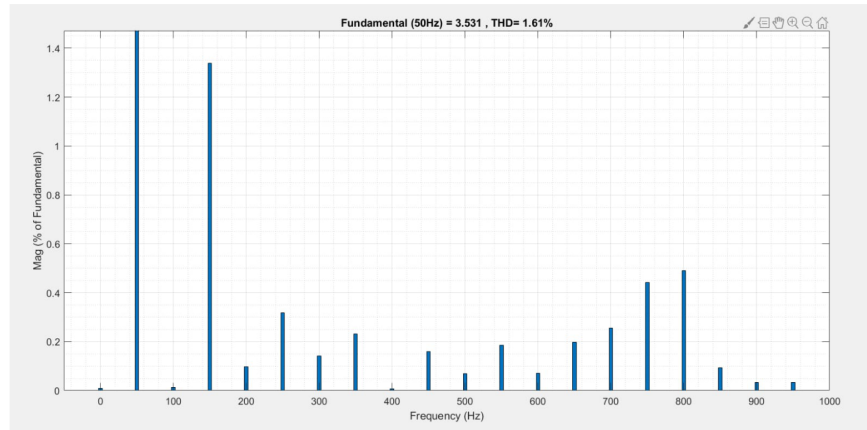


Figure 4.6: FFT spectrum of output current showing a 50 Hz fundamental peak amplitude of 3.531 A and a total harmonic distortion of 1.61%.

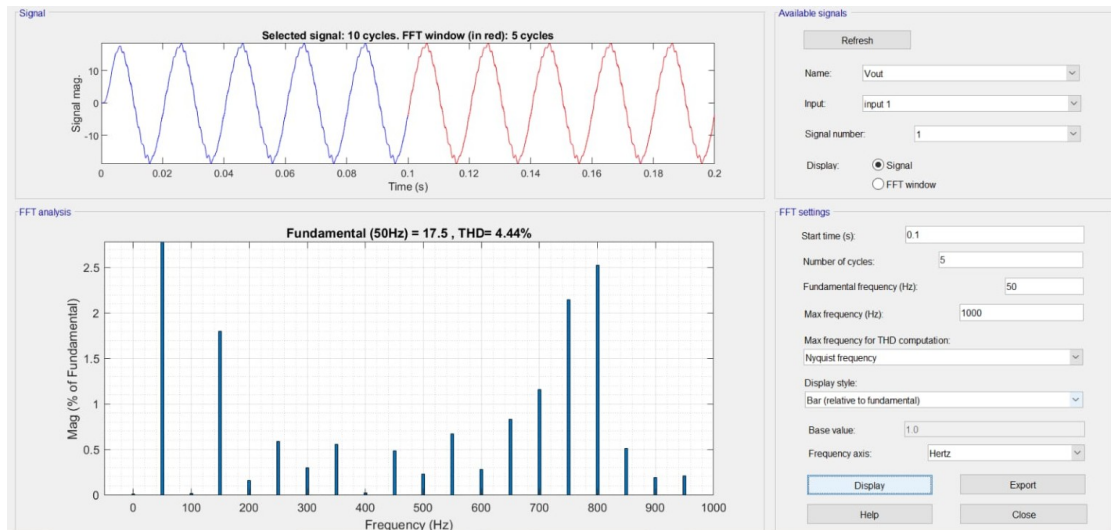


Figure 4.7: FFT-analysis view for the output-voltage signal, showing the selected ten-cycle waveform, the five-cycle steady-state FFT window highlighted in red, and the corresponding harmonic spectrum used to obtain a 50 Hz fundamental peak amplitude of 17.5 V and a voltage THD of 4.44%.

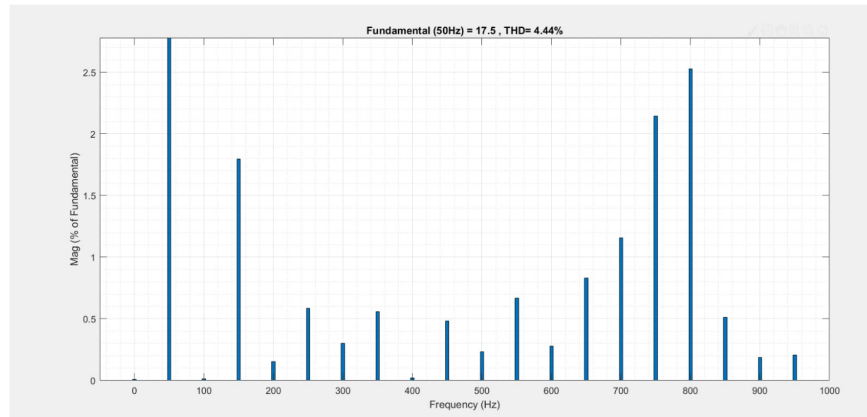


Figure 4.8: FFT spectrum of output voltage showing a 50 Hz fundamental peak amplitude of 17.5 V and a total harmonic distortion of 4.44%.

The FFT tool reports fundamental peak amplitudes. Thus, $17.5/\sqrt{2} \approx 12.375$ V RMS and $3.531/\sqrt{2} \approx 2.497$ A RMS, consistent with the load-side RMS values reported in the results chapter. The lower current THD is expected because the filter and the 5 mH load inductance oppose rapid current change. The voltage waveform is measured at a point where residual PWM and resonance components may remain.

4.9 Performance-Enhancement Measures

4.9.1 Optimising Dead Time

The first enhancement is to use the minimum dead time that safely prevents overlap. The process is empirical:

1. begin with a conservative dead time;
2. observe high-side and low-side gate-source voltages;
3. verify that the outgoing device is off before the incoming gate rises;
4. check drain current and switching-node ringing; and
5. reduce dead time gradually while monitoring temperature and distortion.

A fixed 1 μ s value is appropriate for initial testing but may not be optimal.

4.9.2 Gate-Resistance Tuning

Increasing gate resistance reduces ringing and EMI but increases switching time and loss. Decreasing resistance speeds switching but can create overshoot and driver stress.

Separate turn-on and turn-off resistors with a diode can provide faster turn-off than turn-on. This is useful for reducing shoot-through risk while controlling dv/dt .

4.9.3 Snubber Design

A drain-voltage spike caused by leakage inductance can be estimated from

$$E_\ell = \frac{1}{2} L_\ell I^2. \quad (4.6)$$

An RC snubber absorbs and dissipates part of this energy. Its values should be based on measured ringing frequency and impedance rather than guessed. Excessive snubber capacitance adds switching loss every cycle.

4.9.4 DC-Link Decoupling

A large electrolytic capacitor handles low-frequency energy, while a small film or ceramic capacitor close to the bridge handles high-frequency current. The loop comprising DC capacitor, MOSFETs and transformer primary should have minimum area. Poor layout can produce voltage overshoot even when the average DC bus is only 20 V.

4.9.5 Transformer Improvements

HFT performance can be improved by:

- ensuring equal positive and negative volt-seconds;
- increasing magnetising inductance without excessive turns;
- reducing leakage through winding arrangement;
- using appropriate conductor strands for AC resistance;
- maintaining insulation and creepage; and
- verifying core temperature and flux density.

A small air gap is generally not required for a transformer intended to transfer AC power without storing significant energy, unless needed to control magnetising inductance or avoid saturation from bias.

4.9.6 Filter Damping

The selected LC filter provides strong carrier attenuation but can resonate near 549 Hz. A damping resistor in series with the capacitor can reduce peaking. The resistor

dissipates power, so active damping through feedback is preferable for a high-efficiency design. For the frozen open-loop model, load damping and parasitic resistance are sufficient to obtain the final THD.

4.9.7 Closed-Loop Output Regulation

A future controller can compare output RMS voltage with a reference and adjust m_a . A slow outer loop regulates RMS amplitude, while a faster inner loop can control filter-inductor current. In the synchronous frame this is straightforward for three-phase systems; in a single-phase system, a proportional-resonant controller can provide high gain at 50 Hz without a coordinate transformation. A proportional-resonant controller is

$$G_{PR}(s) = K_p + \frac{2K_r\omega_c s}{s^2 + 2\omega_c s + \omega_o^2}, \quad (4.7)$$

where K_r sets resonant gain and ω_c provides finite bandwidth. Such a controller can reduce steady-state sinusoidal tracking error and compensate DC-link variation.

4.9.8 Active Input Rectification

Replacing the diode bridge with an active rectifier would improve input-current waveform and allow bidirectional power. It would also add four switches, two drivers, current sensing and a DC-link controller. Because the current dissertation is focused on a reproducible low-voltage chain, this enhancement is reserved for future work.

4.10 Switching-Frequency Trade-Off

Increasing f_s reduces transformer volt-seconds per cycle and moves PWM harmonics farther from the fundamental. It can therefore reduce magnetic size and simplify filtering. However, switching and gate-drive losses generally increase with frequency. Core loss also increases as a power of frequency.

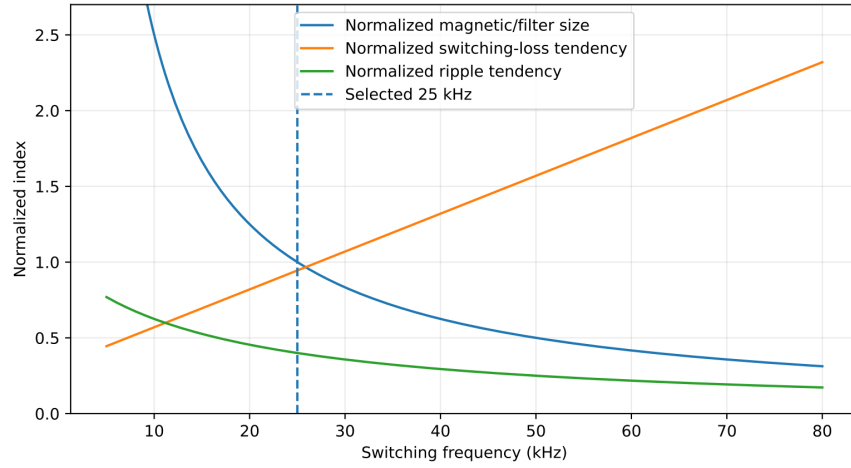


Figure 4.9: Normalised switching-frequency trade-off and the selected 25 kHz operating point.

The selected 25 kHz point is a compromise. It is sufficiently above audio and line frequency to use ferrite magnetics and a compact filter, but low enough for IRF3205 switching with ordinary IR2110 drivers. Raising the frequency substantially would require improved layout, lower-charge devices, faster drivers and a revised core-loss calculation.

4.11 Efficiency Enhancement

The overall efficiency can be written as

$$\eta_{\text{overall}} = \eta_1 \eta_2 \eta_3. \quad (4.8)$$

The logarithmic sensitivity to a stage is

$$\frac{\partial \ln \eta_{\text{overall}}}{\partial \ln \eta_k} = 1. \quad (4.9)$$

Thus, a relative improvement in any stage produces the same relative improvement in the product. In absolute terms, the rectifier has the largest frozen loss and therefore offers the greatest immediate opportunity. Replacing the diode bridge with synchronous rectification could reduce forward-drop loss, but would increase control complexity. The high-frequency stage already has the highest efficiency, 98.3887 %. Its small loss suggests that the unity-ratio HFT and bridge are well matched at the frozen operating point. The output stage has lower efficiency because it includes inverter switching, filter resistance and reactive current.

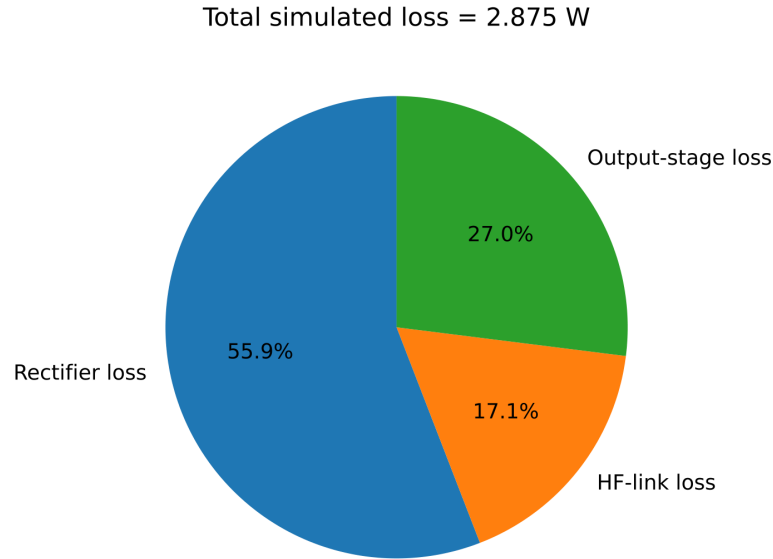


Figure 4.10: Distribution of total simulated loss among the three conversion stages.

4.12 Robustness and Fault Handling

4.12.1 Shoot-Through Protection

Firmware dead time is necessary but not sufficient. A hardware interlock can prevent HIN and LIN from being high simultaneously. Gate-source pull-down resistors prevent floating gates. The gate-driver supply should include local decoupling, and undervoltage lockout must be respected.

4.12.2 Transformer Saturation Protection

A current sensor in the primary can detect a rising current caused by saturation or a secondary fault. The controller should disable the bridge when current exceeds a threshold. A DC-blocking capacitor can prevent steady DC but may introduce resonance and is not a substitute for balanced gating.

4.12.3 Output Short Circuit

The output inverter and filter can deliver high current into a short. Current limiting should be implemented in hardware or in a sufficiently fast control loop. A fuse provides backup isolation. Initial tests should use a low-current bench supply.

4.12.4 Thermal Shutdown

Temperature sensors on MOSFET heatsinks and the HFT can provide a slow protective limit. The allowed temperature should be based on component ratings and insulation class. Simulation efficiency alone does not establish thermal safety.

4.13 Simulation Validation Checks

The following checks are used before accepting a run:

1. gate signals are complementary and include dead time;
2. transformer primary voltage has zero average;
3. primary and secondary square waves are consistent with the 1:1 ratio;
4. the DC-link averages match the frozen values;
5. output voltage and current reach periodic steady state;
6. input and output power are averaged over an integer number of cycles;
7. stage efficiencies reproduce the overall value; and
8. FFT settings reproduce 4.44 % and 1.61 %.

4.14 Chapter Summary

This chapter described the controller, gate timing, simulation structure and measures used to improve performance. Deterministic timer operation, symmetrical HFT excitation, controlled dead time and SPWM are central to the design. The 25 kHz frequency balances magnetic size and switching loss. The LC filter and inductive load explain the final THD relationship. The next chapter presents the numerical results, verifies the power balance and discusses the limitations and future development of the design.

Results, Discussion, Conclusions and Future Scope

5.1 Introduction

This chapter presents the final frozen results of the single-phase SST model. The analysis follows the power path from the AC source to the RL load and checks voltage conversion, current transfer, stage efficiency, total loss and harmonic distortion. The aim is to establish a transparent numerical record rather than to introduce a second set of ratings or alternative simulation values. The principal results are repeated in Table 5.1. All subsequent calculations use these values without modification.

Table 5.1: Final frozen simulation and performance results.

Quantity	Final value	Interpretation
Input power	32.1855 W	Average power entering the conversion chain
Output power	29.3109 W	Average real power delivered to the RL load
Primary DC voltage	20.4614 V	Average after the input rectifier
Primary DC current	1.4927 A	Average entering the high-frequency link
Isolated DC voltage	20.2200 V	Average after the HFT-side conversion
Isolated DC current	1.4859 A	Average entering the output stage
Rectifier-stage efficiency	95.0097%	Input AC to primary DC link
High-frequency-stage efficiency	98.3887%	Primary DC link to isolated DC link
Output-stage efficiency	97.4214%	Isolated DC link to filtered RL load
Overall efficiency	91.07%	Product of the three stage efficiencies
Output-voltage THD	4.44%	Residual voltage distortion after filtering
Output-current THD	1.61%	Load-current distortion

5.2 Input and Rectifier Performance

The 15.5 V RMS source has an ideal peak of 21.92 V. After two frozen 0.2 V diode drops, the approximate rectified peak is 21.52 V. The final average primary DC voltage is 20.4614 V, or about 95.1% of the approximate no-load peak. This is reasonable for

a loaded capacitor-input rectifier.

$$P_{1,\text{out}} = (32.1855)(0.950097) = 30.57935 \text{ W.} \quad (5.1)$$

The corresponding loss is

$$P_{1,\text{loss}} = 32.1855 - 30.57935 = 1.60615 \text{ W.} \quad (5.2)$$

This is the largest of the three stage losses. The bridge diode drops, pulsed source current, capacitor ESR and upstream resistance all contribute. The result indicates that synchronous or active rectification offers the clearest path to a higher overall efficiency, although it would add switches and control.

The product of the separately averaged primary DC voltage and current is

$$V_{\text{DC1}} I_{\text{DC1}} = (20.4614)(1.4927) = 30.5427 \text{ W.} \quad (5.3)$$

The small difference from 30.57935 W is consistent with the distinction between the average of instantaneous power and the product of separately averaged voltage and current.

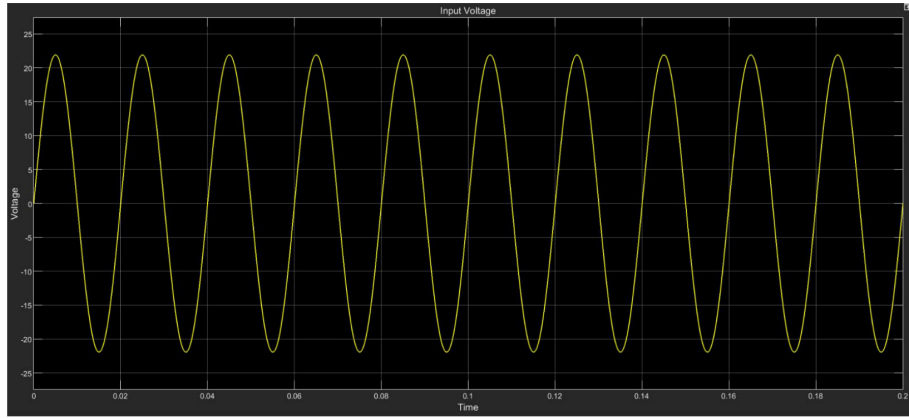


Figure 5.1: Input-voltage waveform of the 15.5 V RMS, 50 Hz source used in simulation.

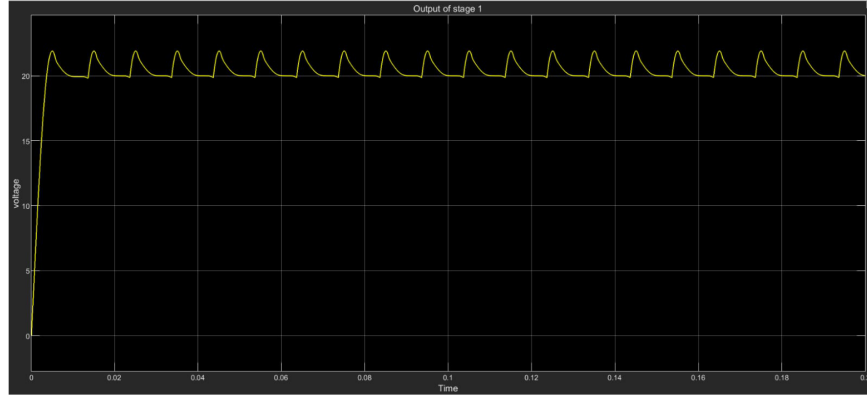


Figure 5.2: Stage 1 output waveform showing the primary DC-link voltage around 20 V.

5.3 High-Frequency Isolation-Stage Performance

The high-frequency stage transfers power from the primary DC link through the MOSFET bridge, 1:1 ferrite HFT and secondary conversion stage. Its efficiency is 98.3887 %, the highest of the three stages. The output power of this stage is

$$P_{2,\text{out}} = (30.57935)(0.983887) = 30.08662 \text{ W.} \quad (5.4)$$

and its loss is

$$P_{2,\text{loss}} = 30.57935 - 30.08662 = 0.49273 \text{ W.} \quad (5.5)$$

The isolated DC voltage falls from 20.4614 V to 20.2200 V, a reduction of 0.2414 V. The current falls from 1.4927 A to 1.4859 A, a reduction of 0.0068 A. These small changes are consistent with the high stage efficiency. The average isolated-side product is

$$V_{\text{DC2}}I_{\text{DC2}} = (20.2200)(1.4859) = 30.045 \text{ W.} \quad (5.6)$$

Its proximity to the efficiency-derived stage power confirms that the frozen average values are internally consistent. The high efficiency should not be interpreted as proof that every hardware HFT will perform similarly. Core material, actual flux density, winding resistance, leakage spikes, gate timing and temperature can change the loss. The simulation result represents the selected model and operating point.

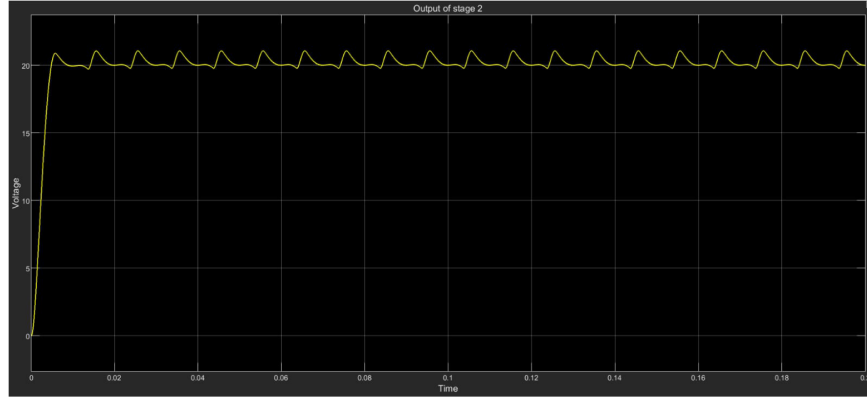


Figure 5.3: Stage 2 output waveform showing the isolated DC-link voltage near 20.22 V.

5.4 Output-Stage Performance

The output stage converts the isolated DC link into a 50 Hz waveform using SPWM, then filters it with 4.2 mH and 20 μ F. Its efficiency is 97.4214 %. The final output power is

$$P_{\text{out}} = (30.08662)(0.974214) = 29.31081 \text{ W} \approx 29.3109 \text{ W}. \quad (5.7)$$

which rounds to 29.3109 W. The stage loss is

$$P_{3,\text{loss}} = 30.08662 - 29.3109 = 0.77572 \text{ W}. \quad (5.8)$$

This loss includes MOSFET conduction and switching, gate-drive energy, filter winding resistance, capacitor ESR and reactive-current circulation. For the 4.7 Ω resistive part of the load, the output current inferred from real power is

$$I_{\text{o,rms}} = \sqrt{\frac{P_{\text{o}}}{R_{\text{L}}}} = \sqrt{\frac{29.3109}{4.7}} = 2.497 \text{ A}. \quad (5.9)$$

With the 5 mH load inductance, the impedance magnitude is approximately 4.956 Ω , so the load voltage magnitude is approximately 12.38 V RMS. This is below the ideal SPWM estimate because of filter reactance, load phase angle and non-ideal converter behaviour.

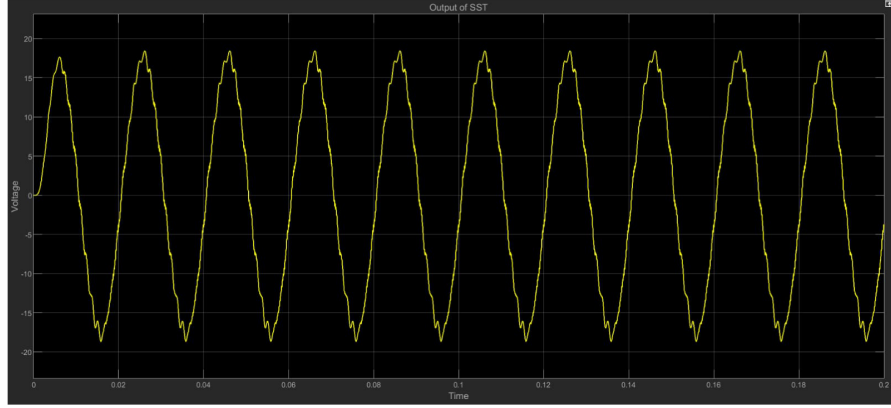


Figure 5.4: Final SST output-voltage waveform measured at the filtered load terminals.

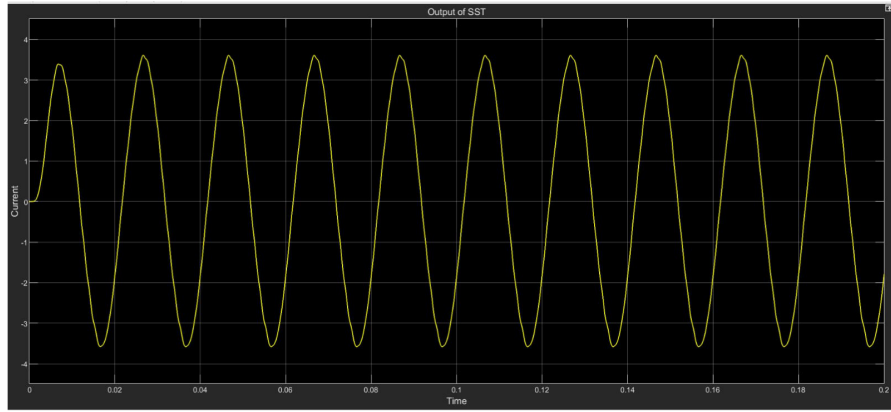


Figure 5.5: Final SST output-current waveform through the RL load.

5.5 Power-Balance Verification

The overall loss is

$$P_{\text{loss,total}} = P_{\text{in}} - P_{\text{out}} = 32.1855 - 29.3109 = 2.8746 \text{ W.} \quad (5.10)$$

The overall efficiency calculated directly from input and output power is

$$\eta_P = \frac{29.3109}{32.1855} \times 100\% = 91.07\%. \quad (5.11)$$

The product of the stage efficiencies is

$$\eta_{\Pi} = (0.950097)(0.983887)(0.974214) = 0.910684 \approx 91.07\%. \quad (5.12)$$

which also rounds to 91.07 %. The agreement verifies the frozen power chain.

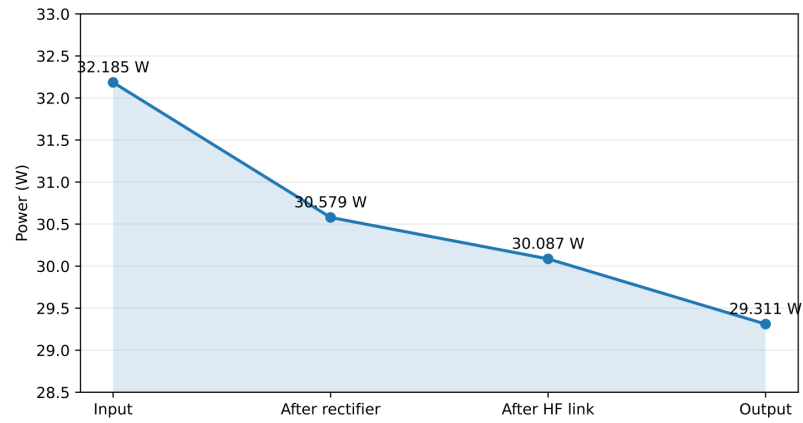


Figure 5.6: Power flow through the three-stage low-voltage conversion chain.

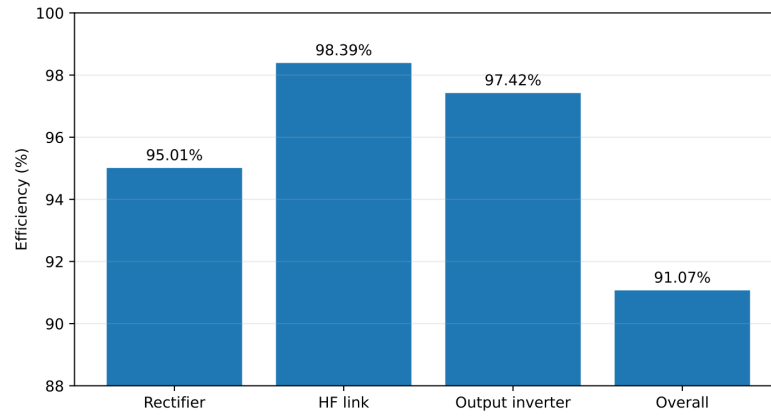


Figure 5.7: Stage and overall efficiencies.

The results illustrate an important property of cascaded conversion: high individual efficiencies do not add; they multiply. The average stage efficiency is approximately 96.94 %, but the overall result is 91.07 %. A dissertation that reports only individual stage values can therefore give an overly optimistic impression.

5.6 Harmonic Results

5.6.1 Output-Voltage THD

The final output-voltage THD is 4.44 %. This result is below the 5 % design target commonly used for a clean laboratory waveform and below the 8 % total voltage distortion limit associated with low-voltage systems in IEEE 519 guidance [12]. The standard is applied at a point of common coupling and contains conditions beyond

this simulation; the comparison is therefore indicative, not a compliance claim. The voltage spectrum contains residual low-order components from DC-link ripple, dead time and modulation non-idealities, as well as high-frequency components associated with PWM. The LC filter strongly attenuates the 25 kHz carrier, but it does not make the output mathematically ideal. A THD of 4.44 % is a credible compromise for an open-loop low-cost design.

5.6.2 Output-Current THD

The final current THD is 1.61 %, substantially lower than the voltage THD. The series load inductance and filter inductor attenuate high-frequency current components because their impedance increases with frequency. The current waveform therefore remains smoother even when small voltage harmonics are present. This result demonstrates why voltage and current THD should both be reported. A low current THD reduces additional copper loss, torque ripple in an inductive machine and current stress in the converter. Voltage THD remains important for insulation, sensitive loads and core loss.

5.6.3 Relationship Between THD and Filter Values

The filter resonance of approximately 549 Hz lies between the fundamental and the switching frequency. At the carrier, the filter inductor reactance is approximately 659.7 Ω and the capacitor reactance is approximately 0.318 Ω . These values explain the strong attenuation of switching-frequency voltage at the load. The filter is not selected solely to minimise THD. Increasing inductance would reduce ripple but increase copper loss, size and fundamental voltage drop. Increasing capacitance would lower resonance and divert more ripple but increase reactive current and start-up stress. The final THD results show that the frozen 4.2 mH/20 μ F choice achieves acceptable quality without an excessively large passive network.

5.7 Discussion of Performance Enhancement

5.7.1 Rectifier Stage

The rectifier accounts for roughly 55.9 % of the total loss. Replacing the diode bridge with low-resistance synchronous switches could reduce conduction loss. An active rectifier could also improve input current quality and regulate the DC bus. The disadvantages are greater control complexity and the need for current sensing.

5.7.2 High-Frequency Link

The isolation stage contributes the smallest loss. Its performance can be preserved by maintaining symmetrical drive, low leakage, short bridge loops and adequate magnetising inductance. The main hardware risk is not average efficiency but switching overshoot or core saturation caused by construction and timing errors.

5.7.3 Output Stage

The output stage offers improvement through unipolar SPWM, better dead-time compensation, lower-resistance filter winding and closed-loop control. A proportional-resonant controller could reduce steady-state amplitude error and selected low-order harmonics. Active damping could allow a sharper filter without resonance.

5.7.4 Semiconductor Upgrade

At the frozen voltage and frequency, IRF3205 devices are sufficient. A switch upgrade should be justified by measured loss. GaN devices could reduce gate charge and switching loss, but their layout and driver requirements are more demanding. SiC devices would be unnecessarily high voltage for a 20 V bus.

5.8 Sensitivity to Operating Conditions

5.8.1 Input-Voltage Variation

Because the input rectifier is passive and the output is open-loop, a change in the 15.5 V source changes both DC links and output voltage. A 5 % input drop can produce a similar output drop unless the modulation index is adjusted. Closed-loop regulation is therefore the most important enhancement for operation with a variable source.

5.8.2 Load Variation

At lighter load, conduction loss falls and efficiency may improve, but fixed switching and gate-drive losses become a larger fraction of output power. At heavier load, voltage drop, copper loss and device temperature increase. The frozen results represent one RL operating point and should not be extrapolated to the full IRF3205 current rating.

5.8.3 Frequency Variation

Lower switching frequency increases transformer flux swing and output ripple. Higher frequency reduces ripple but increases switching and core loss. The 25 kHz setting is therefore part of the design, not an arbitrary simulation parameter.

5.8.4 Modulation Variation

Reducing m_a lowers output voltage and may reduce overmodulation-related distortion, but the load receives less power. Increasing it beyond 0.98 provides little margin and can enter the nonlinear region. A closed loop should remain within a defined modulation limit and use fault handling when the requested voltage cannot be produced.

5.9 Limitations

The main limitations of the work are:

1. the quantitative performance results are based on the simulation model; a preliminary laboratory hardware prototype has been assembled, but it has not yet undergone complete instrumented validation;
2. the rectifier is passive, so input power factor and input-current THD are not actively controlled;
3. the output voltage is primarily open-loop and therefore sensitive to source and load variation;
4. thermal performance is inferred from loss rather than measured junction and core temperature;
5. EMI, common-mode current and insulation ageing are not experimentally evaluated;
6. the HFT turns count and core geometry remain implementation-specific; and
7. the frozen diode and MOSFET parameters are simplified simulation values.

These limitations do not invalidate the performance results. They define the boundary within which the conclusions are valid.

5.10 Laboratory Hardware Prototype

A preliminary low-voltage hardware prototype of the proposed single-phase SST was assembled to support practical implementation and staged validation of the control and power-conversion sections. The setup includes Arduino Mega 2560 controller boards,

breadboard-mounted gate-driver and interface circuitry, perfboard-mounted power devices with heat sinks, an isolation transformer, cooling arrangement and digital measurement instrumentation. The prototype photograph is included to document the physical implementation; the numerical efficiency and THD values reported in this dissertation remain simulation-based until a complete calibrated experimental test is performed.

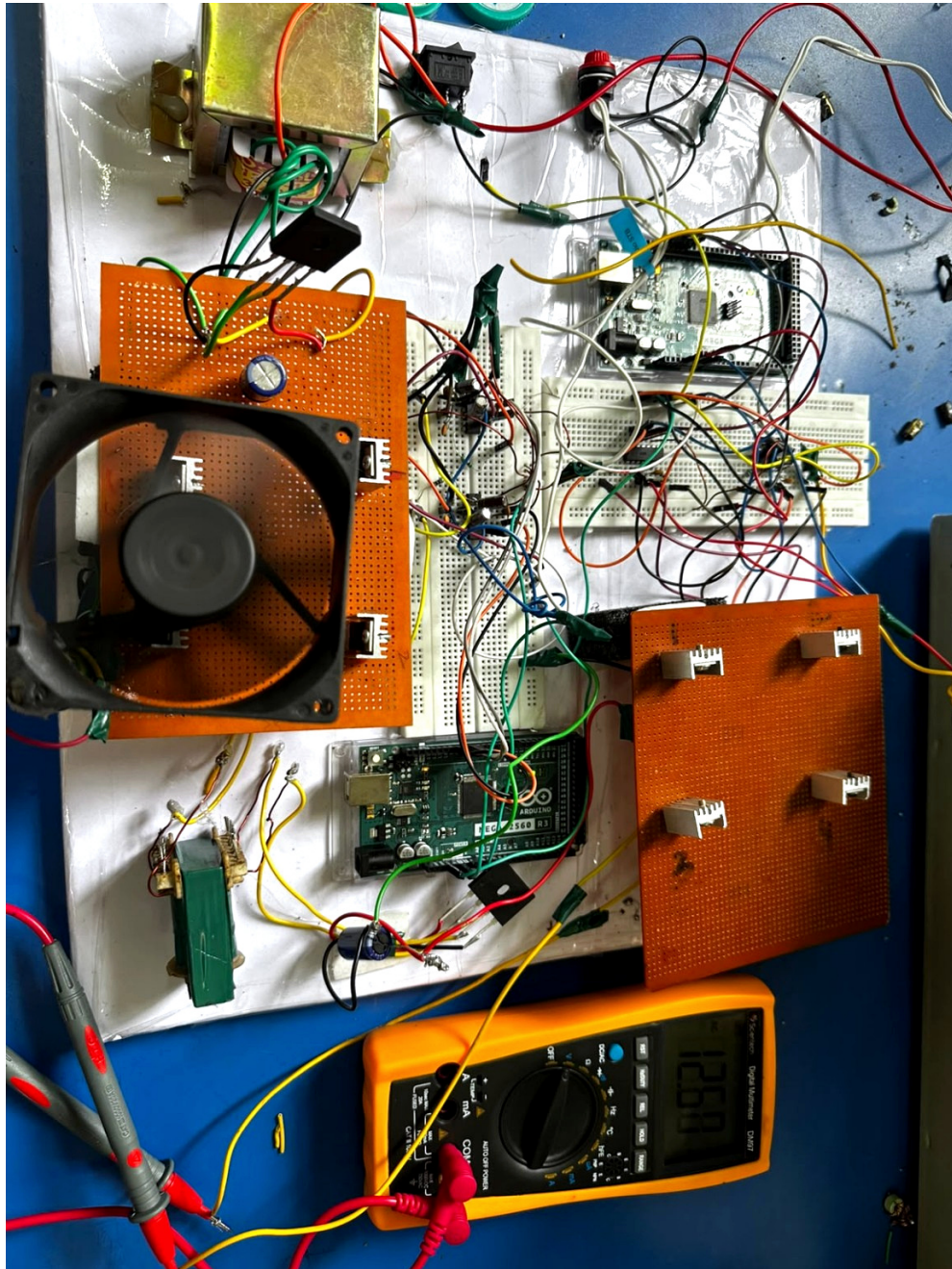


Figure 5.8: Laboratory hardware setup of the low-voltage single-phase SST prototype.

The photograph shows the controller boards, interconnecting gate and measurement wiring, power-stage perfboards, heat sinks, cooling fan, transformer and multimeter

used during preliminary testing. The arrangement is suitable for low-voltage development, although final testing should use shorter power loops, secure insulated terminals, clearly separated control and power wiring, appropriate fusing and current-limited energisation.

5.11 Hardware Validation Plan

A future hardware validation should proceed in stages:

1. verify Arduino gate logic at low frequency with drivers disconnected;
2. verify IR2110 high-side and low-side gate voltages using appropriate differential measurement;
3. test each half bridge from a current-limited DC supply;
4. connect the HFT with the secondary open and measure magnetising current;
5. connect a resistive secondary load and verify the 1:1 voltage transfer;
6. add the secondary rectifier and confirm 20.2200 V near the intended operating point;
7. test the output bridge at a reduced modulation index;
8. connect the LC filter and resistive load;
9. add the 5 mH load inductance; and
10. measure RMS values, instantaneous power, THD and temperature.

Each step should include an emergency shutdown and current limit. A grounded oscilloscope probe must never be connected directly across a floating high-side node unless the measurement arrangement is designed for it.

5.12 Major Contributions

The principal contributions of the dissertation are:

1. a coherent single-phase SST specification based on one frozen parameter set;
2. a complete stage-by-stage electrical model using implementation-oriented components;
3. analytical consistency checks for rectifier voltage, HFT transfer, filter resonance and RL loading;

4. explicit integration of Arduino Mega 2560 control and IR2110 gate-drive requirements;
5. a verified power balance from 32.1855 W to 29.3109 W;
6. final stage efficiencies of 95.0097 %, 98.3887 % and 97.4214 % with overall 91.07 %;
7. final voltage and current THD values of 4.44 % and 1.61 %; and
8. a practical enhancement and hardware-validation roadmap.

5.13 Conclusion

A single-phase low-voltage SST has been designed and modelled using a 15.5 V AC input, full-bridge rectification, a 25 kHz IRF3205 H-bridge, IR2110 gate drivers, a 1:1 ferrite-core HFT, an SPWM output bridge, a 4.2 mH/20 μ F filter and a 4.7 Ω + 5 mH series RL load. The primary and isolated DC averages are 20.4614 V/1.4927 A and 20.2200 V/1.4859 A, respectively. The model converts 32.1855 W of input power into 29.3109 W of load power. The stage efficiencies are 95.0097 %, 98.3887 % and 97.4214 %, and their product gives 91.07 %. The final output-voltage and output-current THD values are 4.44 % and 1.61 %. The lower current THD is consistent with the inductive filtering of the load. The results demonstrate the central SST design trade-off: high-frequency

conversion and active reconstruction provide isolation, controllability and waveform improvement, but every electronic stage contributes loss and implementation complexity. The chosen 25 kHz operating point and LC filter achieve a balanced low-voltage design suitable for laboratory development.

5.14 Future Scope

Future work should prioritise experimental validation and closed-loop operation. The most valuable extensions are:

- hardware construction with a current-limited low-voltage supply;
- measurement of transformer flux, leakage spikes and temperature;
- active or synchronous input rectification;
- unipolar SPWM with timer-based dead-time control;
- proportional-resonant output-voltage regulation;
- active damping of the LC resonance;

- overcurrent, undervoltage and overtemperature shutdown;
- PCB layout designed for short switching loops and controlled grounding;
- comparison of measured and simulated loss; and
- scaling to a higher-power isolated prototype only after the low-voltage system is verified.

A later advanced version could use a bidirectional DAB, isolated current sensing and a DSP controller. Such changes would move the design toward a multifunctional SST while preserving the validated principles of the present work.

5.15 Chapter Summary

The frozen results form a consistent power and harmonic balance. The overall efficiency is independently confirmed by both the input-output ratio and the product

of stage efficiencies. The LC filter and RL load produce an output-current THD below the voltage THD, while the 25 kHz high-frequency link provides unity-ratio isolation with low stage loss. The dissertation therefore provides a complete technical foundation for safe low-voltage hardware validation and future closed-loop improvement.

Chapter A

Frozen Design Data and Engineering Calculations

A.1 Purpose of the Appendix

This appendix records the final numerical values used throughout the dissertation and reconstructs the principal engineering calculations from those values. Its purpose is traceability: every table, graph and conclusion in the main chapters should be consistent with this single frozen data set. Values from earlier medium-voltage or 100 kVA drafts are not design inputs to the present single-phase laboratory model. The system is a low-voltage proof-of-concept SST comprising an uncontrolled input rectifier, a primary high-frequency full bridge, a 1:1 ferrite-core transformer, a secondary rectifier and DC link, and an SPWM output bridge feeding an LC filter and RL load. The frozen parameters represent the final simulation operating point rather than the absolute ratings of the components.

A.2 Master Parameter Register

Table A.1: Frozen values used in the final dissertation.

Parameter	Frozen value	Interpretation
Input source	15.5 V RMS, 50 Hz	Single-phase low-voltage AC supply
Input conversion	Full-bridge diode rectifier	Produces the first DC link
Primary switching bridge	Four IRF3205 MOSFETs	Bipolar high-frequency excitation
Primary bridge drivers	Two IR2110 devices	One half-bridge driver per leg
Controller	Arduino Mega 2560	Timing and PWM logic
High-frequency switching	25 kHz	Frozen carrier/link frequency
Nominal switching period	40 μ s	Reciprocal of 25 kHz
Dead time	1 μ s per transition	Shoot-through prevention interval
Diagonal conduction interval	19 μ s	Complementary bridge interval
High-frequency transformer	Ferrite core, 1:1	Galvanic isolation without voltage scaling
First DC-link voltage	20.4614 V	Time-average value
First DC-link current	1.4927 A	Time-average value
Second DC-link voltage	20.2200 V	Time-average value after isolation
Second DC-link current	1.4859 A	Time-average value after isolation
Output modulation	Sinusoidal PWM	Low-frequency voltage synthesis
Modulation index	0.98	Ratio of sinusoidal reference to carrier amplitude
Output frequency	50 Hz	Fundamental load frequency
Output-filter inductance	4.2 mH	Series filter element
Output-filter capacitance	20 μ F	Shunt filter element
Load resistance	4.7 Ω	Active load component
Load inductance	5 mH	Reactive load component
MOSFET on-resistance parameter	0.02 Ω	Loss-model parameter
Diode forward drop	0.2 V	Rectifier/body-diode loss parameter
Input power	32.1855 W	Frozen average input power
Output power	29.3109 W	Frozen average active output power
Rectifier-stage efficiency	95.0097%	Input rectifier and first DC-link stage
High-frequency-stage efficiency	98.3887%	Bridge, HFT and secondary conversion
Output-stage efficiency	97.4214%	SPWM inverter, filter and load interface
Overall efficiency	91.07%	Product of the three stage efficiencies
Output-voltage THD	4.44%	Fundamental-referenced RMS distortion
Output-current THD	1.61%	Fundamental-referenced RMS distortion

A.3 Input Rectification and First DC Link

For a sinusoidal source of 15.5 V RMS, the ideal peak value is

$$V_{pk} = \sqrt{2}V_{rms} = \sqrt{2}(15.5) = 21.920 \text{ V.} \quad (\text{A.1})$$

A bridge-conduction path contains two diodes. With the frozen forward-drop model, the approximate capacitor-charging ceiling is

$$V_{dc,peak} \approx V_{pk} - 2V_D = 21.920 - 2(0.2) = 21.520 \text{ V.} \quad (\text{A.2})$$

The simulated average of 20.4614 V is lower than this peak because the capacitor

discharges between charging pulses and because source, switch and interconnection impedances are represented in the model. The difference between the estimated charging ceiling and the average is

$$\Delta V_{\text{charge}} = 21.520 - 20.4614 = 1.0586 \text{ V.} \quad (\text{A.3})$$

This quantity should not be interpreted as pure capacitor ripple; it includes the difference between a peak estimate and a time average. The first DC-link average electrical power obtained by direct multiplication is

$$P_{\text{dc1,VI}} = V_{\text{dc1}} I_{\text{dc1}} = (20.4614)(1.4927) = 30.5427 \text{ W.} \quad (\text{A.4})$$

This is close to, but not exactly equal to, the efficiency-derived stage output because independently averaged voltage and current do not generally satisfy $\overline{vi} = \bar{v} \bar{i}$ when correlated ripple is present. The efficiency calculation therefore uses average instantaneous power.

A.4 High-Frequency Bridge Timing

The selected switching frequency gives

$$T_s = \frac{1}{f_s} = \frac{1}{25 \times 10^3} = 40 \text{ } \mu\text{s.} \quad (\text{A.5})$$

The intended sequence is diagonal pair Q1–Q4 for 19 μs , all devices off for 1 μs , diagonal pair Q2–Q3 for 19 μs , and all devices off for 1 μs . The period therefore closes exactly:

$$19 + 1 + 19 + 1 = 40 \text{ } \mu\text{s.} \quad (\text{A.6})$$

The total dead-time fraction of one switching period is

$$D_{\text{dead,total}} = \frac{2(1)}{40} = 0.05 = 5\%. \quad (\text{A.7})$$

Each diagonal pair is commanded for

$$D_{\text{diag}} = \frac{19}{40} = 0.475. \quad (\text{A.8})$$

Table A.2: Frozen high-frequency switching sequence.

Interval	Duration	Devices commanded on	Bridge voltage
1	19 μs	Q1 and Q4	Approximately $+V_{\text{dc1}}$
2	1 μs	None	Freewheel/dead time
3	19 μs	Q2 and Q3	Approximately $-V_{\text{dc1}}$
4	1 μs	None	Freewheel/dead time

A.5 High-Frequency Transformer Ratios

For an ideal transformer,

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}. \quad (\text{A.9})$$

The frozen turns ratio is 1:1, so the secondary square-wave magnitude should follow the primary magnitude apart from winding resistance, leakage inductance, magnetising current and switching drops. The DC-link ratio obtained from the frozen averages is

$$k_{\text{dc}} = \frac{V_{\text{dc2}}}{V_{\text{dc1}}} = \frac{20.2200}{20.4614} = 0.98820. \quad (\text{A.10})$$

The corresponding voltage reduction is

$$\Delta V_{\text{iso}} = 20.4614 - 20.2200 = 0.2414 \text{ V}, \quad (\text{A.11})$$

or approximately 1.18% of the first DC-link average. The current ratio is

$$k_I = \frac{I_{\text{dc2}}}{I_{\text{dc1}}} = \frac{1.4859}{1.4927} = 0.99544. \quad (\text{A.12})$$

The small changes in both DC averages are consistent with a near-unity isolation stage rather than an intentional step-up or step-down converter.

A.6 Output Filter Calculations

For the frozen LC filter,

$$L_f = 4.2 \text{ mH}, \quad C_f = 20 \text{ } \mu\text{F}. \quad (\text{A.13})$$

The undamped natural frequency is

$$f_0 = \frac{1}{2\pi\sqrt{L_f C_f}} = \frac{1}{2\pi\sqrt{(4.2 \times 10^{-3})(20 \times 10^{-6})}} = 549.14 \text{ Hz.} \quad (\text{A.14})$$

The relevant frequency ratios are

$$\frac{f_0}{f_o} = 10.98, \quad \frac{f_s}{f_0} = 45.53, \quad \frac{f_s}{f_o} = 500. \quad (\text{A.15})$$

These ratios provide useful separation between the desired fundamental, the filter corner region and the switching band. At the switching frequency, the individual reactances are

$$X_{L,f_s} = 2\pi f_s L_f = 2\pi(25 \times 10^3)(4.2 \times 10^{-3}) = 659.73 \text{ } \Omega, \quad (\text{A.16})$$

$$X_{C,f_s} = \frac{1}{2\pi f_s C_f} = \frac{1}{2\pi(25 \times 10^3)(20 \times 10^{-6})} = 0.3183 \text{ } \Omega. \quad (\text{A.17})$$

The high series inductive impedance and low shunt capacitive impedance explain the attenuation of carrier components in the load voltage. Practical capacitor-current limits and damping must nevertheless be checked before hardware implementation.

A.7 RL Load Calculations

The load is

$$Z_L = R_L + j\omega L_L, \quad (\text{A.18})$$

where $R_L = 4.7 \text{ } \Omega$, $L_L = 5 \text{ mH}$ and $\omega = 2\pi(50)$. Thus,

$$X_L = 2\pi(50)(5 \times 10^{-3}) = 1.5708 \text{ } \Omega, \quad (\text{A.19})$$

$$|Z_L| = \sqrt{4.7^2 + 1.5708^2} = 4.9555 \text{ } \Omega, \quad (\text{A.20})$$

$$\phi = \tan^{-1}\left(\frac{1.5708}{4.7}\right) = 18.48^\circ, \quad (\text{A.21})$$

$$\text{pf} = \cos \phi = 0.9484 \text{ lagging.} \quad (\text{A.22})$$

Using the frozen active output power and the resistive component,

$$I_{\text{out,rms}} = \sqrt{\frac{P_{\text{out}}}{R_L}} = \sqrt{\frac{29.3109}{4.7}} = 2.4973 \text{ A.} \quad (\text{A.23})$$

The corresponding approximate RMS load voltage is

$$V_{\text{out,rms}} = I_{\text{out,rms}}|Z_L| = (2.4973)(4.9555) = 12.375 \text{ V.} \quad (\text{A.24})$$

The reactive and apparent powers are

$$Q = I_{\text{out,rms}}^2 X_L = 9.796 \text{ var,} \quad (\text{A.25})$$

$$S = V_{\text{out,rms}} I_{\text{out,rms}} = 30.905 \text{ VA.} \quad (\text{A.26})$$

The check $P/S = 29.3109/30.905 = 0.9484$ agrees with the impedance-derived power factor.

A.8 Power-Flow Reconstruction

The output powers after each stage are reconstructed from the frozen efficiencies:

$$P_1 = P_{\text{in}}\eta_1 = (32.1855)(0.950097) = 30.57935 \text{ W.} \quad (\text{A.27})$$

$$P_2 = P_1\eta_2 = (30.57935)(0.983887) = 30.08662 \text{ W.} \quad (\text{A.28})$$

$$P_3 = P_2\eta_3 = (30.08662)(0.974214) = 29.31081 \text{ W.} \quad (\text{A.29})$$

The last value rounds to the frozen 29.3109 W. The overall efficiency is

$$\eta_{\text{overall}} = \eta_1\eta_2\eta_3 = (0.950097)(0.983887)(0.974214) = 0.910684 \approx 91.07\%. \quad (\text{A.30})$$

Table A.3: Stage power and loss reconstruction.

Stage	Input (W)	Output (W)	Loss (W)
Input rectification and DC link	32.1855	30.57935	1.60615
High-frequency isolation	30.57935	30.08662	0.49273
Output inversion and filtering	30.08662	29.3109	0.77572
Complete system	32.1855	29.3109	2.8746

The fraction of total modelled loss associated with each stage is

$$\lambda_1 = \frac{1.60615}{2.8746} = 55.88\%, \quad (\text{A.31})$$

$$\lambda_2 = \frac{0.49273}{2.8746} = 17.14\%, \quad (\text{A.32})$$

$$\lambda_3 = \frac{0.77572}{2.8746} = 26.99\%. \quad (\text{A.33})$$

The rectifier stage is therefore the largest contributor at the frozen operating point. This ranking is specific to the modelled voltage drops, current, switching pattern and selected load.

A.9 Harmonic Data Used for the Final Spectra

The reported THD is calculated as

$$\text{THD} = \frac{\sqrt{\sum_{h=2}^H X_h^2}}{X_1} \times 100\%, \quad (\text{A.34})$$

where X_1 is the RMS fundamental and X_h is the RMS magnitude of harmonic h . The following spectrum tables are normalized to a fundamental of 100%.

Table A.4: Normalized output-voltage harmonic data read from the FFT spectrum.

Order	Magnitude (%)	Order	Magnitude (%)
1	100.0000	11	0.6700
2	0.01436	12	0.2791
3	1.7970	13	0.8320
4	0.1549	14	1.1580
5	0.5856	15	2.1450
6	0.2998	16	2.5260
7	0.5570	17	0.5119
8	0.02047	18	0.1874
9	0.4824	19	0.2063
10	0.2314		

The root-sum-square of the listed non-fundamental voltage components through the 19th harmonic is approximately 4.26%. The FFT tool reports a total voltage THD of 4.44% over its configured analysis bandwidth, which also includes components beyond the displayed 950 Hz range.

Table A.5: Normalized output-current harmonic data read from the FFT spectrum.

Order	Magnitude (%)	Order	Magnitude (%)
0 (DC)	0.009361	10	0.06996
1	100.0000	11	0.18530
2	0.01258	12	0.07115
3	1.3380	13	0.19690
4	0.09780	14	0.25510
5	0.31690	15	0.44222
6	0.14100	16	0.48940
7	0.23100	17	0.09358
8	0.007558	18	0.03236
9	0.16040	19	0.03379

The root-sum-square of the listed non-fundamental current components is approximately 1.61%, reproducing the current THD reported by the FFT tool. The lower current distortion is physically reasonable for an inductive load and series filter because current cannot change instantaneously.

A.10 Consistency Checks

The following checks were applied before freezing the dissertation results:

1. The product of the three stage efficiencies reproduces the overall efficiency.
2. The efficiency-derived final output power reproduces the frozen output power after rounding.
3. The first and second DC-link voltages are compatible with a unity-ratio HFT and modest conversion loss.
4. The 40 μ s period closes exactly using two 19 μ s conduction intervals and two 1 μ s dead times.
5. The LC natural frequency lies well above the 50 Hz fundamental and well below the 25 kHz carrier.
6. The RL-load power factor, active power, reactive power and apparent power satisfy the standard AC power triangle.
7. The current harmonic table reproduces the reported current THD, while the voltage table records the displayed harmonics and the FFT tool supplies the total THD over its full configured bandwidth.

A.11 Limits of Numerical Precision

The number of displayed decimal places does not imply equal experimental certainty. Values such as 95.0097% and 20.4614 V are retained because they are the frozen outputs of the final model. A physical prototype would require measurement-uncertainty analysis, sensor calibration, oscilloscope bandwidth, probe loading, temperature tracking and component-tolerance analysis. For interpretation, efficiencies should normally be discussed to approximately two decimal places and voltages and currents to a precision supported by the instruments.

Chapter B

Essential Control and Hardware Notes

This appendix retains the implementation details needed to reproduce the low-voltage prototype safely; it does not replace a reviewed schematic, component data sheets or laboratory safety procedures.

B.1 Control Signal Allocation

The Arduino Mega provides four independent commands so both devices in each half-bridge can be forced off during dead time.

Table B.1: Arduino Mega signal assignment for the primary bridge.

Pin	Logic	Driver input	Controlled switch
6	HIN1	Driver 1 HIN	Q1, high side of leg A
7	LIN1	Driver 1 LIN	Q2, low side of leg A
8	HIN2	Driver 2 HIN	Q3, high side of leg B
9	LIN2	Driver 2 LIN	Q4, low side of leg B

Table B.2: Frozen bipolar bridge switching sequence.

HIN1	LIN1	HIN2	LIN2	Duration	State
1	0	0	1	19 μ s	Q1+Q4, positive primary voltage
0	0	0	0	1 μ s	Dead time
0	1	1	0	19 μ s	Q2+Q3, negative primary voltage
0	0	0	0	1 μ s	Dead time

HIN and LIN of the same IR2110 must never be high together. Verify all logic outputs with an oscilloscope before connecting the drivers or bridge supply; hardware timers are preferred for deterministic dead time.

B.2 IR2110 and Gate Connections

Table B.3: Essential IR2110 functional connections for each bridge leg.

Terminal	Connection	Implementation note
VCC	Regulated gate-drive supply, typically 12 V	Decouple locally to COM
COM	Low-side power reference	Use a short, low-inductance return
HIN / LIN	Arduino logic commands	Never overlap commands of the same leg
LO	Gate resistor to low-side MOSFET	Add a gate-to-source pull-down
VS	Half-bridge switching node	Floating reference of the high-side channel
HO	Gate resistor to high-side MOSFET	Measure relative to VS, not controller ground
VB	Bootstrap positive node	Connect bootstrap capacitor between VB and VS
Bootstrap diode	VCC to VB, correct polarity	Charges when VS is near COM

Place the bootstrap capacitor close to the driver and refresh it through the switching sequence. Each MOSFET uses a $10\ \Omega$ gate resistor and a gate-to-source pull-down. Measure high-side gate voltage as $V_{HO} - V_S$ with differential or isolated equipment; a grounded probe can short a floating bridge node.

B.3 Safe Commissioning Sequence

The prototype should be energised in the sequence below.

Steps 1–5	Steps 6–10
1. Inspect polarity, MOSFET orientation, bridge wiring and solder joints.	6. Connect the HFT with its secondary open; monitor no-load current and temperature.
2. Power only the Arduino; confirm all outputs remain low during reset.	7. Add the secondary rectifier and light load; confirm polarity and V_{DC2} .
3. Verify the $40\ \mu\text{s}$ period, $19\ \mu\text{s}$ conduction intervals and $1\ \mu\text{s}$ dead times.	8. Enable the output bridge at reduced modulation before adding the LC filter and RL load.
4. Power the IR2110 supply without the DC bus; verify VCC, LO-to-COM and HO-to-VS waveforms.	9. Increase voltage gradually while recording both DC links, input/output power and temperatures.
5. Apply a current-limited low DC voltage and confirm bridge-midpoint switching before connecting the HFT.	10. Acquire FFT data after steady state over an integer number of 50 Hz cycles.

Any overcurrent, DC-link overvoltage, driver undervoltage, overtemperature, invalid complementary state, controller reset or emergency-stop command should force all gate signals low. Fault removal should require a deliberate reset rather than automatic restart.

Bibliography

- [1] X. She, A. Q. Huang, and R. Burgos, “Review of solid-state transformer technologies and their application in power distribution systems,” *IEEE Journal of Emerging and Selected Topics in Power Electronics*, vol. 1, no. 3, pp. 186–198, 2013.
- [2] M. A. Hannan, P. J. Ker, M. S. H. Lipu, Z. H. Choi, M. S. A. Rahman, K. M. Muttaqi, and F. Blaabjerg, “State of the art of solid-state transformers: Advanced topologies, implementation issues, recent progress and improvements,” *IEEE Access*, vol. 8, pp. 19113–19132, 2020.
- [3] J. E. Huber and J. W. Kolar, “Solid-state transformers: On the origins and evolution of key concepts,” *IEEE Industrial Electronics Magazine*, vol. 10, no. 3, pp. 19–28, 2016.
- [4] T. Guillod, F. Krismer, R. Farber, C. M. Franck, and J. W. Kolar, “Protection of MV/LV solid-state transformers in the distribution grid,” in *Annual Conference of the IEEE Industrial Electronics Society*, 2015, pp. 3531–3538.
- [5] A. Q. Huang, M. L. Crow, G. T. Heydt, J. P. Zheng, and S. J. Dale, “The future renewable electric energy delivery and management system: The energy internet,” *Proceedings of the IEEE*, vol. 99, no. 1, pp. 133–148, 2011.
- [6] S. Falcones, X. Mao, and R. Ayyanar, “Topology comparison for solid state transformer implementation,” in *IEEE Power and Energy Society General Meeting*, 2010, pp. 1–8.
- [7] Q. Zhu, L. Wang, D. Chen, L. Zhang, and A. Q. Huang, “Design and implementation of a 7.2 kV single-stage AC-AC solid state transformer based on current source series resonant converter and 15 kV SiC MOSFET,” in *IEEE Energy Conversion Congress and Exposition*, 2017, pp. 1288–1295.
- [8] N. Mohan, T. M. Undeland, and W. P. Robbins, *Power Electronics: Converters, Applications, and Design*, 3rd ed. John Wiley & Sons, 2003.
- [9] R. W. Erickson and D. Maksimovic, *Fundamentals of Power Electronics*, 3rd ed. Springer, 2020.
- [10] H. Qin and J. W. Kimball, “AC-AC dual active bridge converter for solid state transformer,” in *IEEE Energy Conversion Congress and Exposition*, 2009, pp. 3039–3044.
- [11] H. Akagi, “Multilevel converters: Fundamental circuits and systems,” *Proceedings of the IEEE*, vol. 105, no. 11, pp. 2048–2065, 2017.

- [12] *IEEE Recommended Practice and Requirements for Harmonic Control in Electric Power Systems*, IEEE Std 519-2014, 2014.
- [13] H. Shadfar, M. Ghorbani Pashakolaei, and A. Akbari Foroud, “Solid-state transformers: An overview of the concept, topology, and its applications in the smart grid,” *International Transactions on Electrical Energy Systems*, vol. 31, no. 9, p. e12996, 2021.
- [14] A. F. Nardoto, A. E. A. Amorim, L. F. Encarnacao, W. M. Santos, E. J. Bueno, and D. M. Blanco, “Model predictive control for solid state transformer,” *Electric Power Systems Research*, vol. 225, p. 109838, 2023.
- [15] B. Avdeev, A. Vyngra, and S. Chernyi, “Improving the electricity quality by means of a single-phase three-stage solid-state transformer,” *Energies*, vol. 13, no. 18, p. 4789, 2020.
- [16] D. R. Chandran, S. Kumar, and D. Sanath, “Solid state transformers: A comprehensive review of technology, topologies, applications, research gaps, and future directions,” *Journal of Power and Energy Engineering*, vol. 13, no. 6, pp. 30–64, 2025.
- [17] M. E. Adabi, J. A. Martinez-Velasco, and S. Alepuz, “Modeling and simulation of a MMC-based solid-state transformer,” *Electrical Engineering*, vol. 99, pp. 1341–1354, 2017.
- [18] S. Uttar Kumar, D. Agrawal, R. Agrawal, and A. Kumar, “Modeling and analysis of solid-state transformer using MATLAB Simulink,” in *Proceedings of the 2024 IEEE 3rd International Conference on Electrical Power and Energy Systems (ICEPES)*, Bhopal, India, Jun. 21–22, 2024, pp. 1–5.
- [19] C. S. Dessbesell, S. A. Bock, and J. M. Lenz, “Analysis of solid-state transformer features through modeling and control simulation,” presented at IEEE Conference (unpublished, provided as PDF), 2025.
- [20] Yu, “Comparative analysis of SiC and GaN: Third-generation semiconductor materials,” *Highlights in Science, Engineering and Technology*, vol. 81, pp. 484–490, 2024.